

Alicia in Wonderland

The Red & White Queens

"Why, sometimes I've believed as many as six impossible things before breakfast."

The White Queen, in Lewis Carroll, *Through the Looking-Glass* (1871), chapter 5

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).
Cover: the WONDER square (DICTO): WONDER across the top and down the side, each letter once in every row and column, each on a shape of the book: W the White Queen and R the Red Queen (their shadows), O Alicia, N Alice, D Humpty Dumpty, E the White Rabbit; each letter in the colour of its partner.

Opening

Four-dimensional space cannot be entered, but it can be seen: by its shadows, cast into our space as a lamp casts the shadow of a wire frame onto a wall, and by its slices, cut through it as a loaf is cut. This book follows two regular solids of four dimensions through their shadows and slices: the 600-cell, built from 600 tetrahedra, which DICTO named the White Queen, and its polar partner the 120-cell, built from 120 dodecahedra, the Red Queen. Before them come the pieces that lead to them, and after them the finding that ties the whole book together: the five pyramids cut from the centres of the Platonic solids, which Aristotle said fill space, are exactly the cores of the four-dimensional regular solids, seen in perspective.

Alicia and Alice

The title joins two Victorians who never, as far as we know, met. Charles Lutwidge Dodgson (1832–1898) taught mathematics at Christ Church, Oxford, from 1855 to 1881 and wrote as Lewis Carroll. *Alice's Adventures in Wonderland* appeared in 1865, and *Through the Looking-Glass*, where the Red and White Queens move over a chessboard country and everything in the glass goes the other way, in 1871. Dodgson's mathematics was exact and old-fashioned by choice: his *Condensation of Determinants* (1866) gave a method of exact computation still used, and *Euclid and His Modern Rivals* (1879) defended proved geometry against looser textbooks. Late in life, in *Sylvie and Bruno Concluded* (1893), he described Fortunatus's Purse, three handkerchiefs sewn together so that whatever is inside the purse is outside it: an object that cannot be made in our space, his nearest approach to a higher dimension.

Alicia Boole (1860–1940) was born in Cork, the third daughter of the logician George Boole, who died when she was four, and of Mary Everest Boole. She was five when *Alice* appeared and eleven at the *Looking-Glass*. In the 1880s her brother-in-law Charles Howard Hinton, the writer on the fourth dimension, gave her his sets of small wooden cubes, made to train the mind to picture the tesseract. With no formal training she learned to see four-dimensional space so clearly that she found the six regular four-dimensional solids for herself, and the word *polytope* for them is credited to her. After her marriage to Walter Stott in 1890 she learned in 1895 of the work of Pieter Hendrik Schoute at Groningen; he persuaded her to publish, and her paper of 1900, "On certain series of sections of the regular four-dimensional hypersolids", gave the three-dimensional slices of the 600-cell and the 120-cell, drawn in coloured plans and built as card models. Two sets of her 600-cell models are kept at Groningen, which made her an honorary doctor in 1914; from about 1930 she worked with H. S. M. Coxeter until her death.

So the White Queen's slices were first seen by Alicia, two years after Dodgson's death, and this book honours both: Carroll's names are given to the shapes found here (Humpty Dumpty, the White Rabbit, the Mad Hatter, the March Hare, the Lion and the Unicorn, and more), and the slice at the centre of her own series, the 84-corner middle slice of the 600-cell cut cell first, is named Alicia. Chapter 5 states exactly which slices are hers and which are new: her series is cell first, through the layers of corners; the cuts in the gaps between the layers, the other three directions and the shadows are not in it.

Where it started: DICTO's radial cell projection

The idea came while DICTO was building twelve dodecahedra round a thirteenth. In our space they cannot close: gaps open between them. He pictured rays coming out from the centre and passing out through those gaps, gaps that should close in the fourth dimension; and in the 120-cell, the Red Queen of Chapter 6, they do: there every dodecahedron is ringed by twelve others with no gap at all. From that picture came a way to build.

DICTO came to the 600-cell by building it. In Polyhedraverse, one of his apps, he assembled it cell by cell in our space, reflecting each tetrahedron across a face of the last and projecting the result radially, a construction he

called radial cell projection (RCP) and made before he knew of Coxeter's reflection construction: an independent rediscovery of the classical method, combined with a Schlegel-type projection, not a new construction. The model showed the 600-cell's tetrahedra standing in shells round one corner, and the shells became the fifteen Cleo shells of Chapter 5. The app keeps the RCP model side by side with the exact one.

What this book will show

By the end of the book six things will have been shown, each with proof or exact computation.

(1) **Three small pieces close the gaps of icosahedra.** Thirteen icosahedra round one leave gaps that AXE, CLEO and FUJI fill exactly; AXE and CLEO are tiles of Mosseri and Sadoc, who made them to describe icosahedral order in materials, and FUJI is new (Chapter 1).

(2) **Aristotle's pyramids.** Of the five face pyramids of the Platonic solids, exactly the cube's alone, the tetrahedron's with the octahedron's, or these three together fill space; the icosahedron's, the Cleo-Icos, fills space only with a partner from outside the family (Chapter 2).

(3) **Face to face, the pyramids keep to their solids.** Only the tetrahedron's and the octahedron's pyramids share a face-to-face network, the octet; glued together, DICTO's pieces make no new solid with equilateral faces up to four pieces (Chapter 3, first published here).

(4) **The shadows of four dimensions give new deltahedra.** The triangle-faced shadows and slices of the 5-, 16-, 24- and 600-cell include variants of the convex deltahedra that join differently, and new deltahedra rebuilt with every edge equal (Chapter 4).

(5) **The White Queen, complete.** The 600-cell has exactly 62 slice patterns between its centre and its tips in its four main directions, twelve made only of triangles; its shadows resolve into exact halves, and its 600 cells stand in fifteen shells round one corner, the innermost a core of twenty Cleo-Icos (Chapter 5).

(6) **Aristotle's five pyramids are the cores of four dimensions.** Each of the five face pyramids is, with exactly its proportions, the shadow of the core of a four-dimensional regular solid, in polar pairs; the Red Queen 120-cell, built exactly from the White Queen, holds the dodecahedral pyramid at her core (Chapter 6).

Every solid in the book is exact in numbers of the form $a + b\sqrt{5}$ (or with $\sqrt{2}$ and $\sqrt{3}$ where the cube and octahedron enter), and every count and volume is checked exactly. Each chapter is one paper, kept word for word in its theorems and proofs, with the paper and DICTO DOI it comes from; the sources checked for every chapter are listed together at the end.

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Part I. The lead-up



Chapter 1. AXE, CLEO and FUJI

From paper #19, DICTO DOI 10.5281/zenodo.23351291

The gap pieces of the icosahedron: the three pieces that close our thirteen-icosahedron cluster into one larger icosahedron, and where they recur



Figure 1.1. (a) AXE, (b) CLEO and (c) FUJI, at one scale.

Abstract.

Regular icosahedra cannot fill space: two of them meeting along an edge leave a gap of 83.62° around it. Our thirteen-icosahedron cluster, the **Lotus seed**, closes every gap with just three small pieces, **AXE**, **CLEO** and **FUJI**. Together with whole icosahedra they fill an icosahedron of edge φ^2 exactly, in 113 pieces, or in 63 pieces when built as we first built it: a dodecahedron, 12 J11 and then 30 AXE and 20 FUJI.

- **AXE and CLEO are two of the Mosseri–Sadoc tiles**, scaled by $1/\tau$. Sadoc and Mosseri made their tiles to describe icosahedral order in real materials. Here the same tiles arrive on their own, as the gap pieces of a cluster built from whole icosahedra.
- **FUJI, a hexagonal pyramid, is new.** We prove that it is not a union of Mosseri–Sadoc tiles at its own scale or larger. One proof uses the Dehn invariant and the other the angles of its hexagonal base.
- **The three pieces recur throughout our work.** The Dodecahedral Chain passes through the great stellated dodecahedron and the excavated dodecahedron. The dodecahedron is a small icosahedron with 12 caps, 30 AXE and 20 CLEO. The Chain of Karma repeats the Lotus seed at every scale, and the Stella-Corona grows the pieces into a network.

Every volume, angle and count is recomputed from the study data (CHECKS.json). We give printable nets so the Lotus seed can be built by hand.

1. Introduction

Twelve spheres fit around one, and the twelve centres are the corners of an icosahedron. Frank showed that this thirteen-atom cluster is more stable than the thirteen-atom pieces of the cubic and hexagonal crystals [Frank 1952]. It cannot be continued through space, however: the icosahedra around one another leave gaps that no number of icosahedra can close.

Sadoc and Mosseri described this icosahedral order through the 600-cell, a four-dimensional polytope whose 600 tetrahedral cells fit together perfectly on a curved three-dimensional sphere [Sadoc & Mosseri 1999]. Their work

also gave six tetrahedral tiles with golden edge ratios, the Mosseri–Sadoc tiles, from which icosahedral structures are built in flat space [Mosseri & Sadoc 1982; Kramer 1982; Koca et al. 2020].

We came to the same gaps from the other side, with whole icosahedra in flat space. Our Lotus seed puts 12 icosahedra around a small one and fills what is left [Icosa-13 study]. Two of the three pieces that do the filling turn out to be Mosseri–Sadoc tiles; the third, FUJI, is new. This chapter describes the three pieces (Section 2), the Lotus seed they complete (Section 3), and the solids, chains and networks they build elsewhere (Sections 4 and 5). Section 6 proves FUJI's independence, and Section 7 gives the nets.

Units: the icosahedra have edge 1, and $\varphi = \tau = (1 + \sqrt{5})/2$.

2. Three pieces

Figure 1.1, at the head of the paper, shows the three pieces at one scale, the one at which they fill the Lotus seed.

The names are ours. **AXE** is shaped like an axe head, with its sharp edge as the blade. **CLEO** is named after Cleopatra's Needle, the obelisk in London. **FUJI** has the shape of Mount Fuji, and its colour is that of the fuji flower, Japanese wisteria.

Piece	Shape	Edges	Volume	Credit
AXE	tetrahedron	five of length 1, one of $1/\varphi$	1/12	Mosseri–Sadoc *t6* at scale $1/\tau$
CLEO	triangular pyramid	base equilateral $1/\varphi$, side edges 1	$(\sqrt{5} - 1)/24$	Mosseri–Sadoc *t4* at scale $1/\tau$
FUJI	hexagonal pyramid	base edges 1 and $1/\varphi$ alternating, all six side edges 1	$(1 + 3\sqrt{5})/24$	new

AXE is two equilateral triangles hinged at 41.81° (cos = $\sqrt{5}/3$). Two icosahedra sharing an edge leave 360° − 2 × 138.19° = 83.62° around it, as two gaps of 41.81°, one on each side. An AXE closes each one.

CLEO is a slender pyramid whose four faces meet at 63.43° along its long edges. Placed on a face of an icosahedron of edge $1/\varphi$, twenty CLEOs make the spikes of Kepler's great stellated dodecahedron (Section 4).

FUJI has a hexagonal base with all angles 120° and sides 1 and $1/\varphi$ alternating. The base is cyclic: all six corners lie on a circle of radius $\sqrt{(2/3)}$, which puts the apex at height $1/\sqrt{3}$ with every side edge 1. FUJI sits in the corner where three icosahedra meet, with its apex at a corner of the dodecahedron inside the cluster.

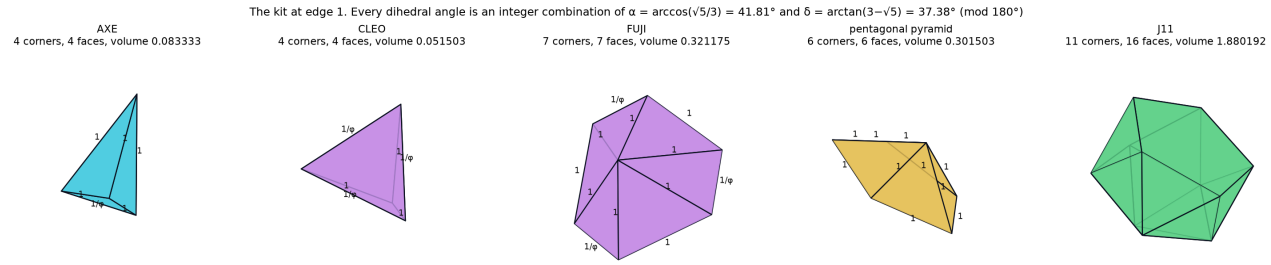


Figure 1.2. The pieces at edge 1, every edge labelled: AXE, CLEO, FUJI, the pentagonal cap J2 and J11 [Chain of Karma].

Two angles. All the dihedral angles of the three pieces are made of two angles, $\alpha = \arccos(\sqrt{5}/3) = 41.81°$ and $\delta = 37.38°$ (FUJI's base angle on its short edges):

Angle	As α, δ	Where
41.81°	α	AXE's hinge; FUJI's base on its long edges
37.38°	δ	FUJI's base on its short edges
63.43°	π − α − 2δ	AXE's short edge; CLEO's side edges
79.19°	α + δ	AXE's other four edges; CLEO's base
116.57°	α + 2δ	the dodecahedron's dihedral angle
142.62°	π − δ	FUJI's side edges

These relations hold to 120 digits (integer-relation search, Icosa-13 study §9). The same two angles reappear in Section 6, where they make the Dehn invariants exact.

3. The Lotus seed

Twelve icosahedra placed on the five-fold axes of a small central icosahedron, parallel to it and slid inward until they meet, share 30 whole edges. Their inner rings are the 12 faces of a dodecahedron of edge 1. R. W. Gray described this arrangement in 2002 and noted that the 12 icosahedra outline a larger one [Gray 2002]; we call the cage **DESHI**. Gray did not fill the gaps; the Lotus seed does.

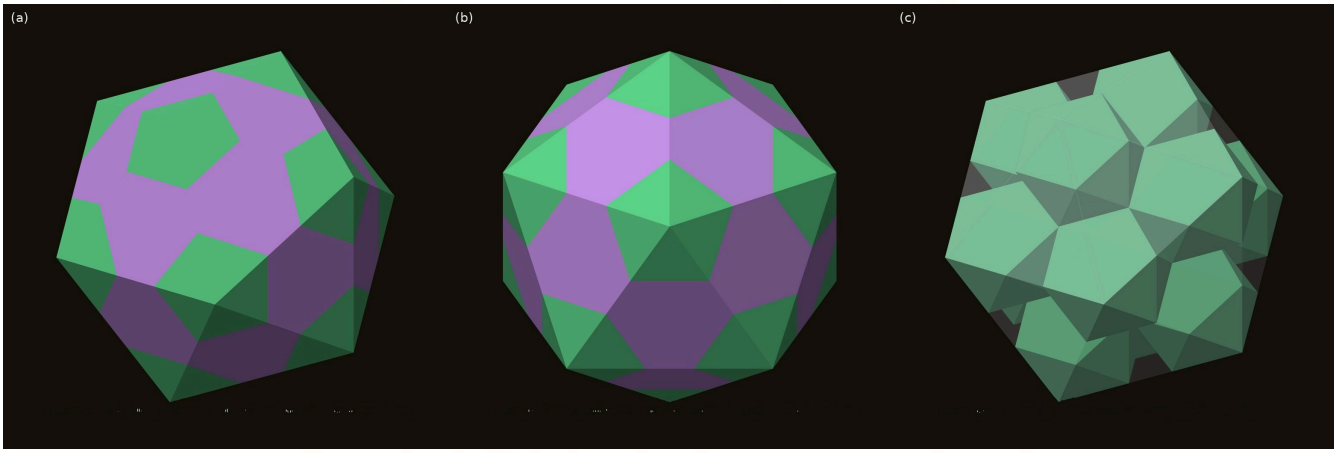


Figure 1.3. The Lotus seed: (a) the filled cluster; (b) down a five-fold axis; (c) its hull, a regular icosahedron of edge φ^2, over the cage of 12 icosahedra.

The filling. Thirty AXE close the outer grooves, thirty more close the inner ones, 20 FUJI fill the outer corners, and 20 CLEO fill the inner corners around a small icosahedron of edge $1/\varphi$:

$I(\varphi^2) = 12\, I(1) + I(1/\varphi) +$ 60 AXE + 20 FUJI + 20 CLEO (113 pieces, no overlaps, volume $(235 + 105\sqrt{5})/12 = \varphi^6$ V(I)).

The result is exactly a regular icosahedron of edge φ^2. Its 20 large faces are each three icosahedron faces and one FUJI base.

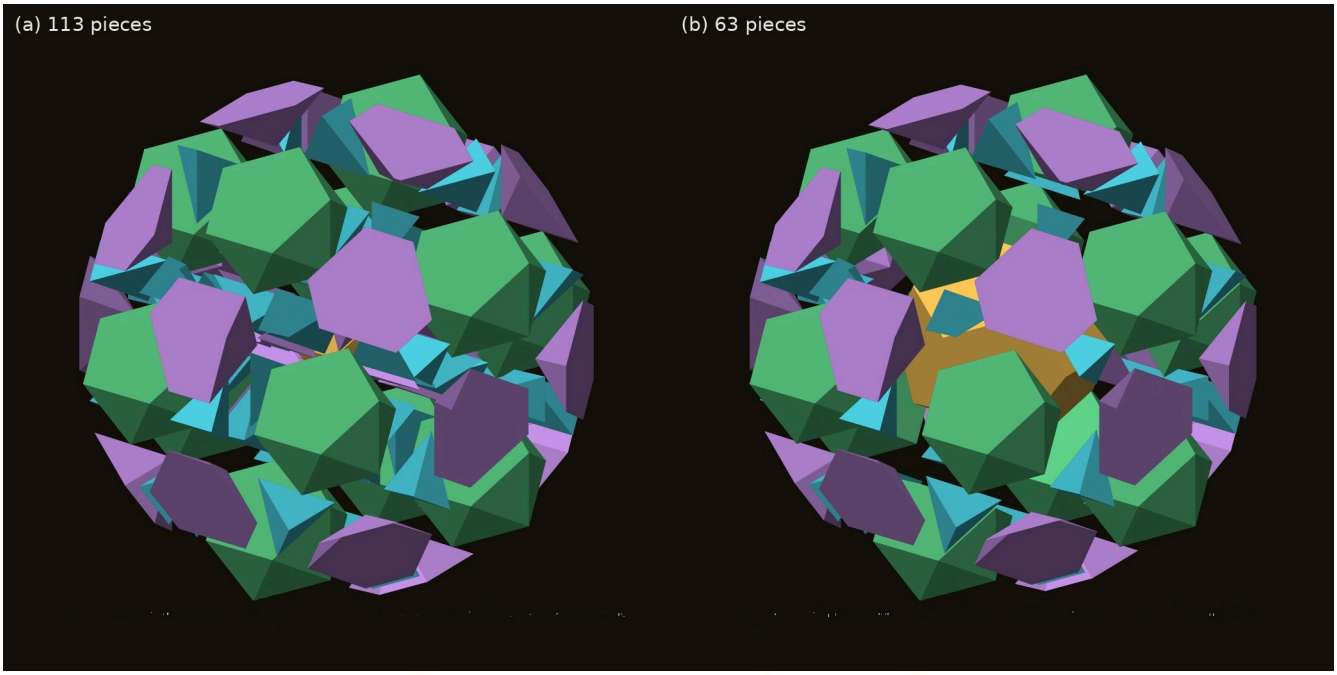


Figure 1.4. The Lotus seed pulled apart: (a) the 113-piece build; (b) our 63-piece build. Green: icosahedra and J11; cyan: AXE; purple: FUJI and CLEO; gold: the centre (a) and the dodecahedron (b).

The 63-piece build. We first made this solid in the Polyhedraverse app on 5 October 2026, while working on a dodecahedron problem, by putting a J11 on every pentagon of a dodecahedron. Each J11, together with the cap hidden inside the dodecahedron, is one of the 12 outer icosahedra, so:

I(φ²) = dodecahedron + 12 J11 + 30 AXE + 20 FUJI (63 pieces).

The small centre, the inner AXE and the CLEOs merge into the single dodecahedron. Two standard solids and two gap pieces make it the easiest way to build the Lotus seed by hand (Section 7). The same move, a J11 on every unit pentagon, applied to an icosidodecahedron instead of a dodecahedron gives our **UNITY** [Icosa-13 study].

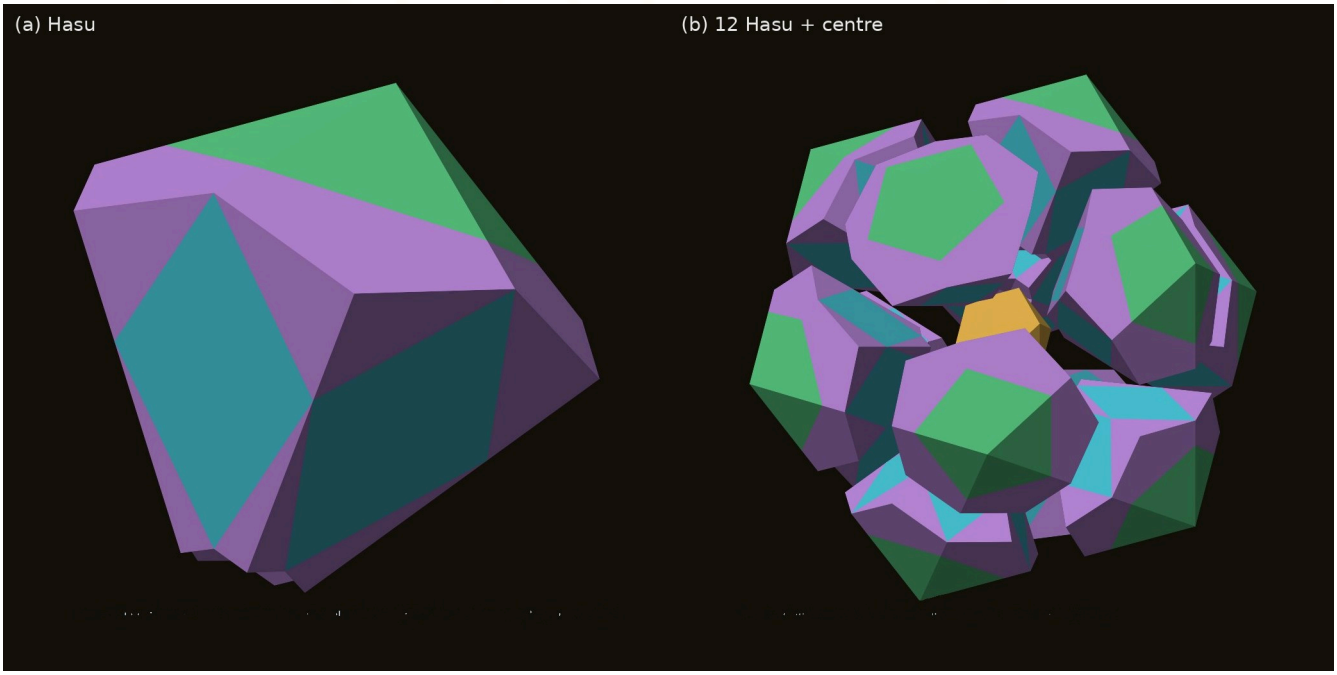


Figure 1.5. (a) **Hasu**, the finned unit: an icosahedron with its share of the gap pieces (21 parts, 45 faces). (b) Twelve Hasu and the centre rebuild the Lotus seed, and each lifts straight out along its axis.

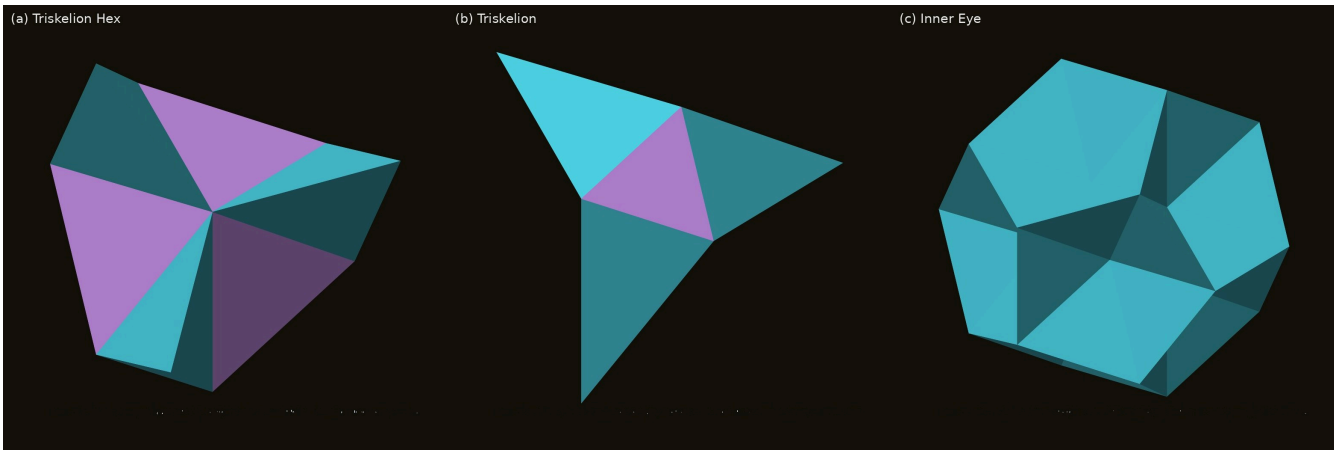


Figure 1.6. The pieces around each dodecahedron corner and the core. (a) **Triskelion Hex** = FUJI + 3 AXE. (b) **Triskelion** = CLEO + 3 AXE. (c) **Inner Eye** = I(1/φ) + 30 AXE + 20 CLEO, the dodecahedron with its 12 caps removed (volume 5φ/2).

The Triskelion has exactly the volume of the pentagonal cap J2, $(5 + \sqrt{5})/24$.

4. The same pieces, other solids

The dodecahedron. Adding the 12 caps to the Inner Eye gives the dodecahedron, so

D(1) = I(1/φ) + 12 J2 + 30 AXE + 20 CLEO.

Built in that order, the stages are classical solids [Capturing the In-betweens]:

- the icosahedron I(1/φ) **+ 20 CLEO is Kepler's great stellated dodecahedron**;
- **+ 30 AXE is the excavated dodecahedron**, a stellation of the icosahedron, on the same 32 corners [Coxeter et al. 1938; Wenninger 1971];
- **+ 12 caps is the dodecahedron.**

This is the **Dodecahedral Chain**: from the icosahedron to its dual through two star solids, with AXE and CLEO doing all the work.

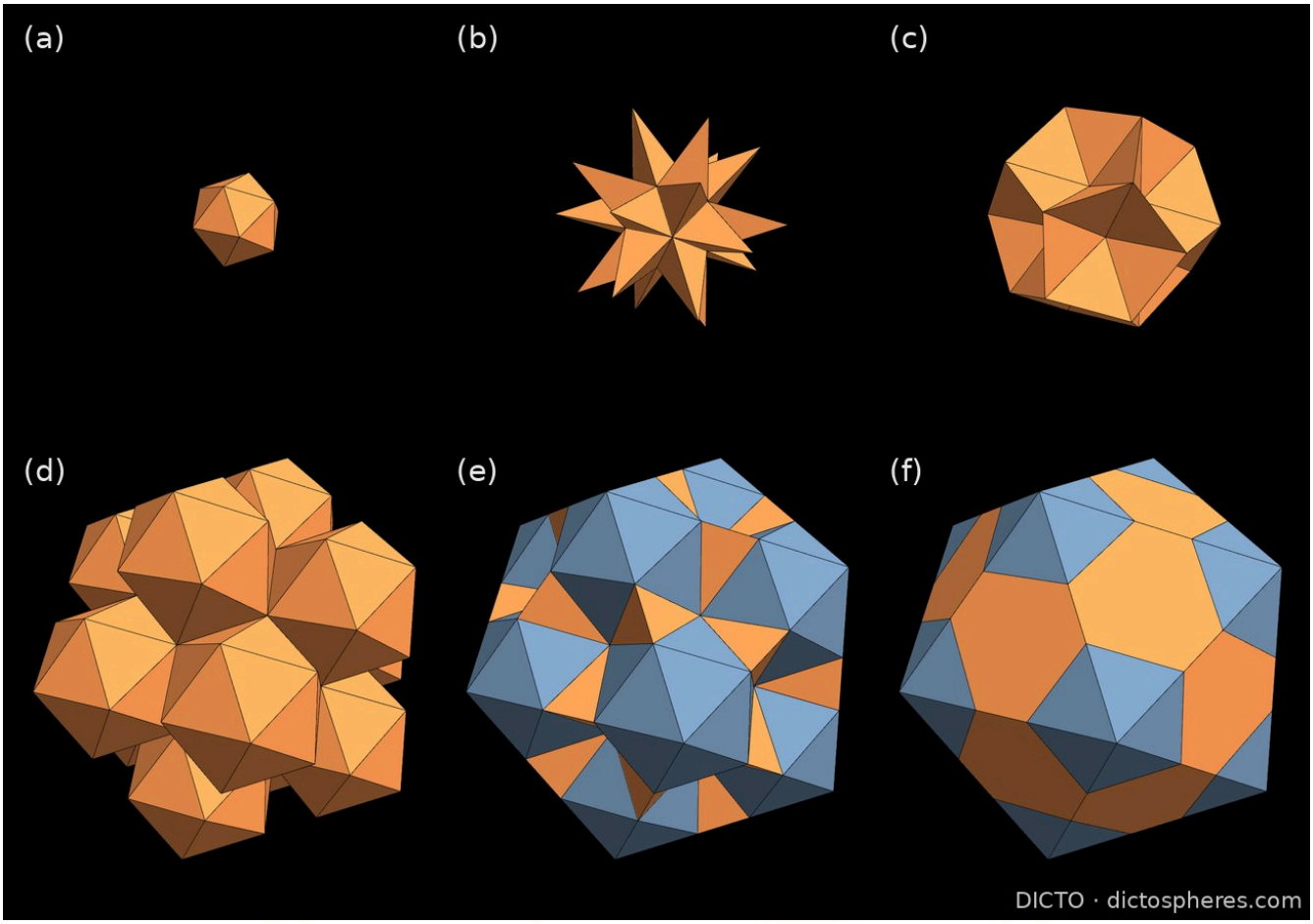


Figure 1.7. The Lotus seed built outward: (a) core icosahedron; (b) + 20 CLEO, the great stellated dodecahedron; (c) + 30 AXE, the excavated dodecahedron; (d) + 12 icosahedra; (e) + 30 AXE; (f) + 20 FUJI, the icosahedron of edge φ^2. Orange: the pieces added at each step.

5. Chains and networks

The Chain of Karma. The Lotus seed is an icosahedron enlarged by φ^2, made of icosahedra and the three pieces. Replacing every icosahedron in it by a Lotus seed, and keeping the gap pieces whole, fills the icosahedra of edge φ^4, φ^6, … exactly. Steps 1 to 3 have 113, 1,569 and 20,497 pieces [Chain of Karma]. AXE, CLEO and FUJI stay whole at each step, so they appear in every size, one step of φ^2 apart. The enlarged pieces cannot be cut back into the small ones (their Dehn invariants forbid it), so the Chain is a hierarchy with gap pieces in every size, rather than a self-similar tiling.

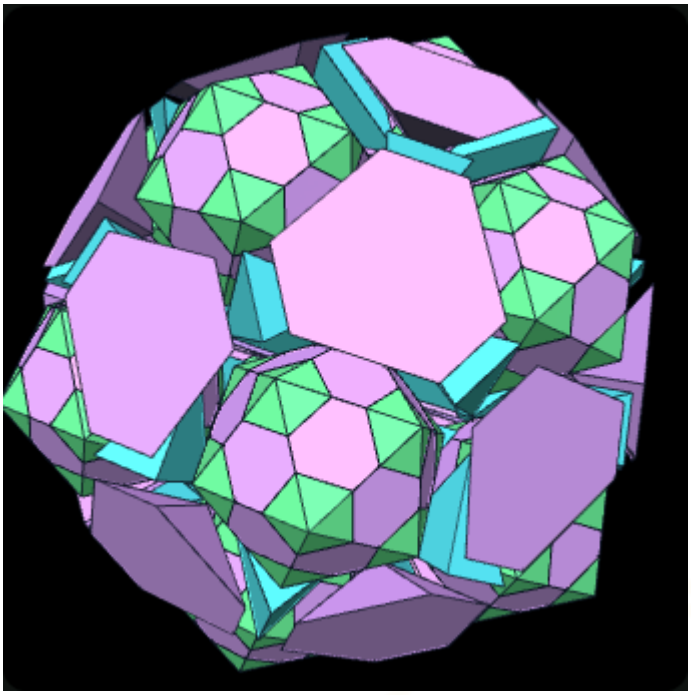


Figure 1.8. The Chain of Karma, step 2: 1,569 pieces filling the icosahedron of edge φ^4.

The Stella-Corona. On an icosidodecahedron we place 20 icosahedra on the triangles and a **King of Pentacles** on each of the 12 pentagons: a J2, 10 AXE, 5 CLEO and a small J2, with a flat pentagram on top. The result is closed, every contact is a whole face, and its volume is $(120 + 47\sqrt{5})/3$. Stella-Coronas that share icosahedra form a chain along a three-fold axis and a diamond network, the first periodic network of icosahedral clusters in this work [Icosa networks study].

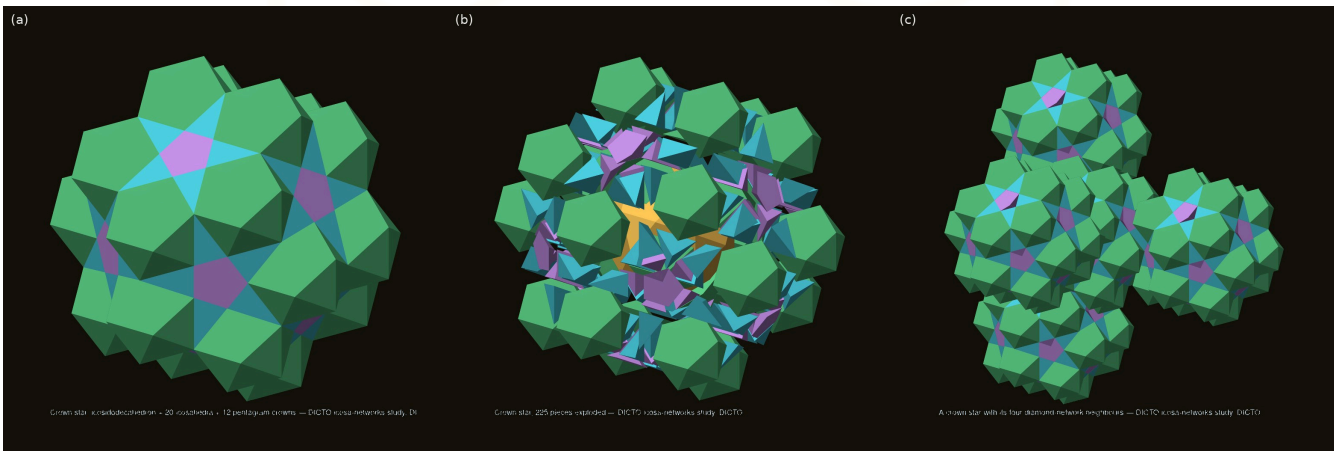


Figure 1.9. The DICTO Stella-Corona: (a) closed; (b) pulled apart; (c) in its diamond network.

6. FUJI stands alone

AXE and CLEO are Mosseri–Sadoc tiles. Is FUJI made of them?

Volume does not decide it. FUJI has exactly the volume of 3 AXE + CLEO + the smallest Mosseri–Sadoc tile *t1* at scale $1/\tau$.

The Dehn invariant does. Two solids can be cut into each other only if they have equal Dehn invariants: the sum over edges of length $\otimes$ dihedral angle, with angles counted modulo rational multiples of π. With every angle written in α and δ, and every length as $a + b\sqrt{5}$:

	1-part (α, δ)	√5-part (α, δ)
FUJI	(3, −15/2)	(0, 3/2)
3 AXE + CLEO + *t1′*	(27/2, 15/2)	(−1/2, 1/2)

The √5-parts differ by (α + 2δ)/2 = (π − arctan 2)/2. Since cos²(arctan 2) = 1/5, and a rational multiple of π with rational cos² can only have cos² equal to 0, 1/4, 1/2, 3/4 or 1, arctan 2 is not a rational multiple of π. So the invariants differ, and no such dissection exists.

No Mosseri–Sadoc mix at this scale or larger. A search over all six tiles at four scales finds other mixes with FUJI's volume and Dehn invariant. The base of FUJI settles them all:

- Each 120° corner of the hexagon must be made as 60° + 60°, from two equilateral tile faces lying in the base. (The Mosseri–Sadoc faces have angles 60°, 36°/72° or 36°/108°.)
- At these scales the equilateral faces have side 1 or 1/τ. So the base must hold a side-1 triangle on each long edge and a side-1/τ triangle on each short edge.
- Those six triangles have a total area of 1.795, more than the hexagon's 1.669. Two of them would overlap, so no such dissection exists.

Mixes of smaller Mosseri–Sadoc tiles remain open. CLEO itself fits inside FUJI, with room to spare: it can grow to 1.023 times its size there.

So FUJI is a genuinely new piece, and the Lotus seed needs it. Koca et al. note that the Mosseri–Sadoc tiles alone cannot inflate the icosahedron past edge τ [Koca et al. 2020]. Taking whole icosahedra as pieces, with FUJI in their corners, the Lotus seed reaches φ², and the Chain of Karma carries on without end.

7. Build it

The nets (nets/axe-cleo-fuji-nets.pdf) print on A4 at 45 mm per unit, giving a Lotus seed with edges of about 118 mm. They contain one dodecahedron, 12 J11, 30 AXE and 20 FUJI, with CLEO for the 113-piece build and the star solids. Each net is checked: every face is congruent to the solid's, every edge is either a fold or a numbered glue pair, and nothing overlaps.

To build: glue a J11 on each face of the dodecahedron. Then fill the 30 grooves between neighbouring J11 with AXE, and the 20 hexagonal dents at the dodecahedron's corners with FUJI. The finished solid is a regular icosahedron.

Chapter 2. Aristotle's Pyramid

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Theorem: of the five face pyramids of the Platonic solids, only the cube's alone and the tetrahedron's with the octahedron's fill space; the icosahedron's, the Cleo-Icos, needs a partner that is not a face pyramid

Abstract. Main theorem. Of the five face pyramids of the Platonic solids (the cone from the centre over one face), a set fills space exactly when it is the cube's pyramid alone, the tetrahedron's and the octahedron's pyramids together, or the union of these two; no other set fills space. Here a set fills space when copies of its pyramids, in any numbers, any positions and mirror images, every kind used and not necessarily face to face, tile space. The proof is a Dehn-type counting map on dihedral angles, made rigorous with the Splitting Theorem of Conway, Radin and Sadun: the dodecahedron's and the icosahedron's pyramids can never occur, and the tetrahedron's occurs exactly when the octahedron's does; the octet truss cut from its centres and the cubic tiling give the tilings. Corollaries: alone only the cube's pyramid fills space, in pairs only the tetrahedron's with the octahedron's, and with equal edges their numbers are asymptotically equal; the icosahedron's pyramid, the Cleo-Icos (DICTO's name), fills space only with a partner that is not a face pyramid, Antonius (#20). The question is DICTO's: Aristotle wrote that among solids only "the pyramid and the cube" fill space (On the Heavens III.8); read as the regular tetrahedron this is false, and DICTO asked whether he could have meant the Cleo-Icos. A section after the theorem collects DICTO's earlier exact fillings by the whole Platonic solids with partners. Every angle identity, the test over all 31 sets and the constructions are checked in exact arithmetic.

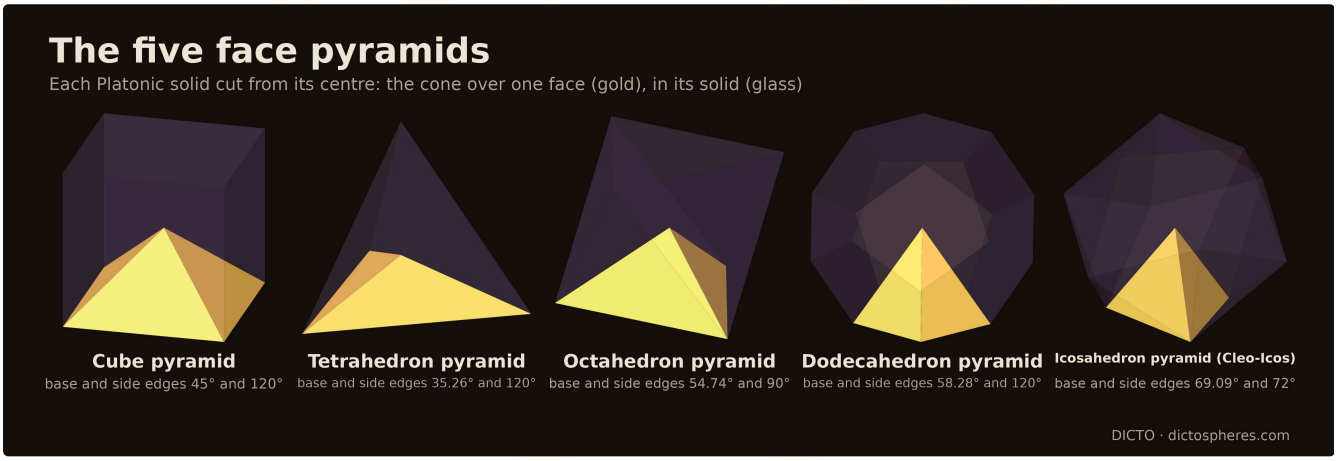


Figure 2.1. The five face pyramids (gold), each shown in its solid (glass) with its tip at the solid's centre and the face at the bottom: cube, tetrahedron, octahedron, dodecahedron and icosahedron (the Cleo-Icos). Under each: the dihedral angles along its base edges and its side edges (Table 1).

1. The theorem

Face pyramids. For a Platonic solid with centre O, the face pyramid over a face F is the convex hull of F and O (Figure 2.1). Its dihedral angle along a base edge is half the solid's dihedral angle, since the plane through O and the edge bisects it; along a side edge (from O to a corner) it is 360°/q, where q faces meet at that corner. We write C, T, O, D and I for the face pyramids of the cube, tetrahedron, octahedron, dodecahedron and icosahedron; I is DICTO's Cleo-Icos, reached from his CLEO piece [2, 3].

Face pyramid	Base edges	Dihedral along a base edge	Side edges	Dihedral along a side edge
cube	4	45° = π/4	4	120° (3 faces at a corner)
tetrahedron	3	α/2 ≈ 35.26°	3	120°
octahedron	3	(π − α)/2 ≈ 54.74°	3	90° (4 faces)
dodecahedron	5	$\theta_D/2$ ≈ 58.28°	5	120°
icosahedron (Cleo-Icos)	3	$\theta_I/2$ ≈ 69.09°	3	72° (5 faces)

Table 1. The five face pyramids. α = arccos(1/3) ≈ 70.53°, θ_D = arccos(−1/√5) and θ_I = arccos(−√5/3) are the dihedral angles of the tetrahedron, dodecahedron and icosahedron; the octahedron's is π − α (its cosine is −1/3) and the cube's π/2.

Filling space. A set U of face pyramids *fills space* if there is a tiling of space whose tiles are congruent copies (mirror images allowed) of pyramids of U, at finitely many fixed sizes, in any numbers and any positions, every kind in U used, with no face-to-face condition.

> *Cut each Platonic solid from its centre into identical pyramids, one on every face. Aristotle wrote that the pyramid fills space; DICTO asked whether he could have meant one of these. The theorem gives the complete answer: only the cube's pyramids fill space on their own, and only the tetrahedron's and the octahedron's do it together. The dodecahedron's and the icosahedron's can never be used at all.*

> **Main theorem.** A non-empty set U of the five face pyramids fills space if and only if U is {C}, {T, O} or {C, T, O}: the cube's pyramid alone, the tetrahedron's and the octahedron's pyramids together, or both of these. More precisely, in every tiling of space by copies of face pyramids, (a) no tile is a D or an I, and (b) T tiles occur if and only if O tiles occur.

2. Proof

> *The proof rests on a simple fact: wherever pieces meet along an edge, their angles round that edge must add up to a full turn.*

Lemma 1 (angles round an edge). Let space be tiled by convex polyhedra, finitely many meeting any bounded set, and let e be an edge of a tile P. For all but finitely many points x of e, the tiles containing x have x either in the relative interior of an edge on the line of e, or in the relative interior of a face whose plane contains that line; the wedge angles at x (the dihedral angles of the tiles with x on an edge) add up to 2π, or to π when one tile has x inside a face (the edge lies on a face).

Proof. Only finitely many tiles meet a neighbourhood of e. Leave out of e the finitely many points that are corners of these tiles, or where an edge of one of them crosses the line of e. At any other point x of e, no tile has x in its interior (it would overlap P), and a tile with x inside an edge has that edge on the line ℓ of e. If a tile T has x inside a face, near x it is a half-space; P lies in the other closed half-space, and so does the line ℓ through x, so the plane contains ℓ. Near x every tile is therefore a wedge or a half-space bounded by planes through ℓ, and in the plane at right angles to ℓ through x they become sectors that cover a disc with disjoint interiors: their angles add up to 2π. A half-space counts π, and there is at most one, since two would leave no room for P. ∎

Lemma 1 is the edge-counting step behind the classical fact that a polyhedron which tiles space has Dehn invariant zero (Debrunner [9]; Lagarias and Moews [10]); we prove it here directly because the tilings may use several kinds of tiles.

Now let h be a map from angles to the rationals that is additive with rational coefficients and has h(π) = 0. At every such point x the values of h at the dihedral angles there add up to h(2π) = 0 or h(π) = 0. Such maps exist with any values on α, θ_D and θ_I, because:

> *The awkward angles of the tetrahedron, dodecahedron and icosahedron are unrelated to one another, so none of them can make up for another.*

Lemma 2. π, α, θ_D and θ_I are linearly independent over the rationals.

Proof. $\cos^2\alpha = 1/9$, $\cos^2\theta_D = 1/5$ and $\cos^2\theta_I = 5/9$, so $\tan\alpha = 2\sqrt{2}$, $\tan\theta_D = -2$ and $\tan\theta_I = -2/\sqrt{5}$: all three are pure geodetic angles in the sense of Conway, Radin and Sadun [4], with tangents rational multiples of $\sqrt{2}$, $\sqrt{1}$ and $\sqrt{5}$, three different square classes. None is a rational multiple of π: if $\theta = (m/n)\pi$ then $e^{i\theta}$ is a root of unity and $2\cos\theta$ an algebraic integer, but $(2\cos\theta)^2$ is 4/9, 4/5 and 20/9, not integers. (Theorem 2 of [4] writes each angle as a rational multiple of π plus their basis angles; this shows the basis part is not zero.) Suppose a rational combination $q_1\alpha + q_2\theta_D + q_3\theta_I$ is a rational multiple of π. By the Splitting Theorem of [4, Theorem 3], so is each part whose tangents are rational multiples of one square root: $q_1\alpha$, $q_2\theta_D$ and $q_3\theta_I$ separately; hence $q_1 = q_2 = q_3 = 0$. ∎

In this basis the cube pyramid's angles are $\pi/4$ and $2\pi/3$, the side edges' $2\pi/3$, $\pi/2$ and $2\pi/5$, all rational multiples of π ($h = 0$); the base edges carry $\frac{1}{2}\alpha$ (T), $\frac{1}{2}\pi - \frac{1}{2}\alpha$ (O), $\frac{1}{2}\theta_D$ (D) and $\frac{1}{2}\theta_I$ (I). This is the argument of #20 for the Cleo-Icos alone [3, §5.6–5.7], carried out for all five and made rigorous with [4].

> *Keep count of one awkward angle at a time. A dodecahedron or icosahedron pyramid leaves an angle nothing can complete; a tetrahedron pyramid's angle can be completed only by an octahedron pyramid's, and the other way round.*

Proof of (a) and (b). (a) Choose $h(\theta_D) = 1$ and $h(\pi) = h(\alpha) = h(\theta_I) = 0$. The base edges of a D carry $h = \frac{1}{2}$, and every other dihedral angle of every pyramid carries 0. At a point x of a base edge of a D, chosen as in Lemma 1, the values of h add up to at least $\frac{1}{2}$, not 0: a contradiction. The same with $h(\theta_I) = 1$ excludes I. (b) Choose $h(\alpha) = 1$, the rest 0: a T's base edges carry $+\frac{1}{2}$, an O's $-\frac{1}{2}$, all other angles 0. Without O, a base edge of a T gives a sum of at least $\frac{1}{2}$; without T, a base edge of an O gives at most $-\frac{1}{2}$. So a set that fills space contains neither D nor I, and contains T exactly when it contains O: it is {C}, {T, O} or {C, T, O}.

> *And these three ways really work: six cube pyramids make a cube, and the classical octet of tetrahedra and octahedra, cut from their centres, gives the other two.*

The three sets fill space. {C}: six cube pyramids make a cube. {T, O}: Maurolico's octet [1] of regular tetrahedra and octahedra, each cut from its centre into its face pyramids. Two tetrahedra on opposite faces of an octahedron make a rhombohedron of 60°, one cell of the face-centred cubic lattice [1]; with lattice vectors (1, 1, 0), (1, 0, 1), (0, 1, 1) the octahedron is 4/3, each tetrahedron 1/3 and the cell 2, and it cuts into 8 octahedron pyramids and 8 tetrahedron pyramids (Figure 2.2, top). Round each edge of the octet, between 2 tetrahedra and 2 octahedra, the pyramids give $4 \times \alpha/2 + 4 \times (\pi - \alpha)/2 = 2\pi$. {C, T, O}: place the octet with its octahedra centred at the integer points of odd coordinate sum (corners at distance 1 along the axes) and its tetrahedra in the unit cubes; then every face pyramid lies in a slab $k \le z \le k + 1$, so the plane $z = 0$ is made of pyramid faces. Keep the octet's pyramids below it and fill the half-space above it with unit cubes cut into cube pyramids. ∎

Verification. pyramid_checks.py checks in exact arithmetic every angle in Table 1 (through $\cos^2$ and the sign of cos), the facts used in Lemma 2, the test (a)–(b) over all 31 non-empty sets of the five pyramids (only {cube}, {tetrahedron, octahedron} and {cube, tetrahedron, octahedron} pass, and with all five allowed D and I are still excluded), the octet cell and its pyramids, and the slabs on a patch of 112 octahedron and 32 tetrahedron pyramids.

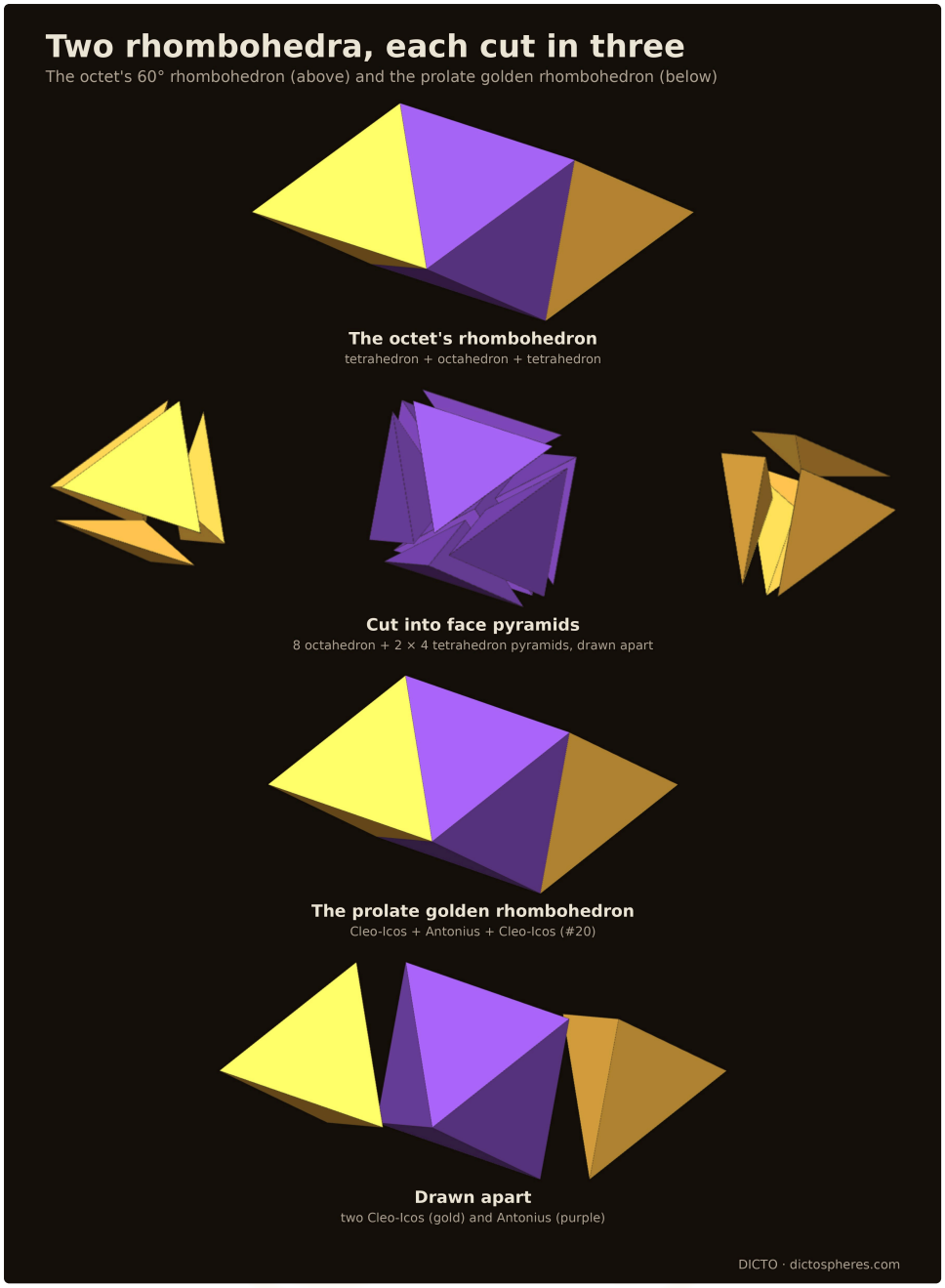


Figure 2.2. Above: one octet cell, the 60° rhombohedron of tetrahedron + octahedron + tetrahedron, whole (gold, purple, gold) and cut into face pyramids: the octahedron into its 8 pyramids (purple) and each tetrahedron into its 4 (gold), drawn slightly apart along their own axes. Below: the prolate golden rhombohedron, Cleo-Icos + Antonius + Cleo-Icos (#20), whole and drawn apart along its long axis. Each panel is scaled to fit.

3. Corollaries

> *Alone, only the cube's pyramid works; in pairs, only the tetrahedron's with the octahedron's.*

Corollary 1 (alone and in pairs). Alone, only the cube pyramid fills space. In pairs, only the tetrahedron and octahedron pyramids do; every other pair fails, the pairs with the cube pyramid because their second piece cannot be used.

> *When tetrahedron and octahedron pyramids fill space together, they come in equal numbers over large regions, as long as their edges are equal.*

Corollary 2 (equal numbers). In any tiling of space by tetrahedron pyramids of edge a and octahedron pyramids of edge b, the numbers $N_T(R)$ and $N_O(R)$ of each inside a ball of radius R satisfy $|a\,N_T(R) - b\,N_O(R)| \le C\,R^2$ for a

constant C; so their numbers are in the ratio b : a in the limit, equal when the edges are equal, as in the octet.

Proof. With h(α) = 1, Lemma 1 gives a sum of 0 at all but finitely many points of each edge. Integrate along all edges inside the ball: Σ over tiles of Σ over their edges of h(angle) × (length inside the ball) = 0. A tetrahedron pyramid wholly inside contributes 3 × a × ½, an octahedron pyramid 3 × b × (–½); the tiles that meet the sphere are at most C′R² in number (bounded size, volume bounded below) and contribute a bounded amount each. ∎

> *The icosahedron's pyramid, the Cleo-Icos, can fill space only with a different kind of partner: DICTO's Antonius, two Cleo-Icos to one Antonius.*

Corollary 3 (the Cleo-Icos needs Antonius). The Cleo-Icos fills space with no set of face pyramids; a partner must be another kind of solid. Antonius is one [3, §5.7]: the three lines from the icosahedron's centre to the corners of a face meet at arccos(1/√5) ≈ 63.43°, the angle of the golden rhombus, so they span a prolate golden rhombohedron, and the planes x + y + z = 1 and 2 (in coordinates along those lines) cut it into a Cleo-Icos at each sharp corner and **Antonius** in the middle, a trigonal antiprism of exactly 4 times the Cleo-Icos's volume (the rhombohedron is 6 times). Golden rhombohedra tile space by a lattice, so **Cleo-Icos and Antonius fill space, two to one** (Figure 2.2, below). In the counting map's terms: along Antonius's 6 triangle edges the angle is $\pi - \theta_I/2 \approx 110.91°$, which with the Cleo-Icos's $\theta_I/2$ makes a flat 180°, and along each of its 6 zigzag side edges 108° (the rhombohedron's 72° edges are the 6 at its sharp tips, which belong to the Cleo-Icos); with $h(\theta_I) = 1$ and edges weighted by length, a Cleo-Icos carries 3 × ½ and Antonius 6 × (–½), –2 times as much, which the two-to-one ratio balances (the Dehn invariant remark of [3]). The same cut of the 60° rhombohedron gives the octet's tetrahedron + octahedron + tetrahedron [3]: in both, the middle third is the partner. (Checked exactly in pyramid_checks.py.)

4. Conjectures

None: every statement above is proved. An open question, not conjectured here, is which tilings by tetrahedron and octahedron pyramids exist besides the octet cut up and its rearrangements.

5. The whole solids and their partners

> *The whole solids can all fill space as well, each with the right partners; the table gathers how.*

DICTO asked, too, how the whole Platonic solids fill space. His earlier exact work answers it for all five:

Solid	Fills space with	Source
cube	alone	classical
tetrahedron, octahedron	each other, two tetrahedra to one octahedron: the tetrahedral–octahedral honeycomb (Maurolico; Fuller's octet truss)	classical; Maurolico (1529) [1]
dodecahedron	Dogstars: dodecahedra on alternate cubes of a cubic lattice, each a cube with six of Euclid's roofs [5]; the hole in each other cube is the Dogstar; the dodecahedra take (5 + √5)/8 ≈ 90.45% of space	tiling and Dogstar: Hart (1996) [6]; exact: The DICTO Sunstar, #42 [7]
icosahedron	DICTO's icosahedral octet: an icosahedron + 6 edge roofs (Admiral Caps) is the regular octahedron, so icosahedra (edge 1) with their roofs in the octet's octahedra, and icosahedra of edge 1/2 with corner and edge caps in its tetrahedra, fill space; the icosahedra take (175 – 75√5)/12 ≈ 60.79%	exact (released finding F21; not a separate paper) [8]; the icosahedron in the octahedron at the golden section is classical (Pacioli; Fuller's Jitterbug)

Table 2. How each whole Platonic solid fills space with partners. The idea of F21 is DICTO's: "the same principle as the dodecahedron and roofs, but with six" [8].

The pattern is the same for both golden solids. Euclid builds the dodecahedron on a cube with a roof on each face [5]; the roofs are what let dodecahedra sit on a cubic lattice, and what is left is the Dogstar. DICTO moved the roofs to the icosahedron: six edge roofs make it an octahedron, which the octet already places. The main theorem says why such help is needed: a solid's face pyramids carry half its dihedral angle along each base

edge, and only for the cube ($\pi/2$) is that angle a rational multiple of π; the tetrahedron's and the octahedron's cancel each other (α and $\pi - \alpha$), while θ_D and θ_I cancel with nothing among the five.

6. Aristotle, briefly

> *What Aristotle wrote, how it was argued over for two thousand years, and where DICTO's question fits.*

"Among surfaces it is agreed that there are three figures which fill the place that contain them—the triangle, the square and the hexagon: among solids only two, the pyramid and the cube" (Aristotle, On the Heavens III.8, 306b, in Guthrie's translation as quoted by Lagarias and Zong [1]). Read as the regular tetrahedron, the claim is wrong: five tetrahedra round an edge leave 7.36° [1]. Lagarias and Zong trace the long correction [1]: Simplicius and Averroës held that twelve regular tetrahedra fill space round a point; others said twenty; Thomas Bradwardine observed that the icosahedron cuts from its centre into twenty congruent tetrahedra but could not decide whether they are regular (they are not: Lagarias and Zong note that their solid angle at the centre is about ten percent larger than a regular tetrahedron's); Paul of Middelburg asserted that no number of regular tetrahedra fills space round a point, arguing from Euclid's Book XIII; and Maurolico (1529) stated that cubes alone, and tetrahedra together with octahedra, fill space. Twenty regular tetrahedra can meet at a point with small gaps [1]: the near miss behind the commentators' twenty.

We do not claim to know what Aristotle meant; Lagarias and Zong take "pyramid" to be the regular tetrahedron [1], the usual reading. DICTO asked whether it could have been Bradwardine's tetrahedron, the Cleo-Icos. Read as a face pyramid, the answer is exact: Aristotle's two figures come out right only as the cube's own pyramid (trivially, since six make the cube) and as the tetrahedron's pyramid taken together with the octahedron's, Maurolico's correction [1] cut from the centres; the Cleo-Icos needs Antonius.

7. Conclusion

> *If Aristotle's pyramid is a face pyramid, his claim is now settled exactly; what he himself meant remains unknown.*

Read as a statement about face pyramids, Aristotle's claim is settled by the main theorem: the cube's pyramid fills space alone, and the only other way is the tetrahedron's and the octahedron's pyramids together, in asymptotically equal numbers when their edges are equal. That pair is the octet truss cut from its centres, the classical fix that Maurolico gave for the regular solids. The icosahedron's pyramid, the Cleo-Icos, cannot join any set of face pyramids; its golden fix is Antonius, with which it fills the prolate golden rhombohedron two to one (#20). For the whole solids the common thread is Euclid's roof: on the cube it makes the dodecahedron that fills space with Dogstars, and moved to the icosahedron as six edge roofs it makes the octahedron that the octet places. What stays open is which other tilings by tetrahedron and octahedron pyramids exist, and what Aristotle himself meant.

8. Prior art, and the status of each result

- **Known, credited:** the history of Aristotle's claim, the near miss, Bradwardine's remark on the icosahedron's twenty tetrahedra and Lagarias and Zong's note that they are not regular, Paul of Middelburg's argument from Euclid's Book XIII, Maurolico's tiling and the 60° rhombohedron of two tetrahedra and an octahedron (Lagarias and Zong [1]); Dehn's angle argument and the geodetic-angle theorems (Conway, Radin and Sadun [4]); the tiling of the cube's face pyramids (six make a cube); the dodecahedron–Dogstar tiling (Hart [6]); the icosahedron in the octahedron at the golden section (Pacioli; the Jitterbug); the Cleo-Icos itself, Mackay's unit tetrahedron and a pyramid of Hypsicles (as credited in [3]).
- **DICTO's earlier work, cited:** the Cleo-Icos and Antonius filling and the counting argument for the Cleo-Icos alone (#20 [3]); the exact Sunstar Lattice (#42 [7]); the icosahedral octet (F21 [8]).

- **Proved here in general:** the main theorem, with Lemmas 1 and 2, and Corollaries 1–3 (Corollary 3 with the filling of [3]). The principle behind it is classical: a polyhedron that tiles space has Dehn invariant zero (Debrunner [9]; Lagarias and Moews [10]); Lemma 1 with the counting map is a self-contained proof of the special case needed here, and what is new is the classification of the face-pyramid sets.
- **Proved for the cases built (exact checks):** the angles of Table 1, the octet cell and its 8 + 8 pyramids, the slabs that let cube pyramids join the octet, the golden rhombohedron's split and Antonius's angles.
- **Observed numerically:** only the PSLQ search of angle_relations.py, which finds no integer relation among π, α, θ_D, θ_I; it is evidence only, Lemma 2 being the proof.

In the sources checked (the full text of the 81 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's 175 link-only entries were not searched) we did not find a classification of the tilings by the face pyramids of the Platonic solids.

Chapter 3. The Five Pyramids: Shared Networks and Deltahedra

First published in this book, from DICTO's study of 27 October 2026 (studies/2026-10-27-five-pyramids).

Abstract. Cut a Platonic solid from its centre to each face and it falls into congruent face pyramids; the icosahedron's is the Cleo-Icos (DICTO's name). Chapter 2 showed which sets of the five face pyramids fill space when they may touch anyhow. DICTO proposed three further questions: which of the five can share a network face to face, which new deltahedra his pieces make when glued together, and how such solids join. We prove that face to face the five pyramids behave exactly as their whole solids: the only network shared by two different face pyramids is the octet cut from its centres, the tetrahedron's pyramid with the octahedron's, and the Cleo-Icos meets another face pyramid only base to base, in finite bipyramids. A search of every gluing of up to four pieces (five for DICTO's pieces alone) finds 772 solids with equilateral faces; every one is a regular solid cut up again or a face-gluing of regular tetrahedra, octahedra and icosahedra, so none is new, and none of them fills space alone.

Aristotle wrote that the pyramid fills space. The five pyramids cut from the centres of the Platonic solids are the natural candidates, and they lead to the four-dimensional solids of Part III, where each of them turns out to be the core of one. This chapter asks how the five behave when they must meet whole face to whole face, and whether DICTO's own pieces, glued together, make new solids with equilateral faces, new deltahedra.

1. The five face pyramids

With the edge of each solid 1, $\alpha = \arccos(1/3)$, $\theta_D = \arccos(-1/\sqrt{5})$ and $\theta_I = \arccos(-\sqrt{5}/3)$, the five pyramids are (exact values, sympy):

Pyramid	Faces	Base-edge dihedral	Side-edge dihedral
Cube (C)	square 1; 4 isosceles (1, √3/2, √3/2)	45°	120°
Tetrahedron (Q)	equilateral 1; 3 isosceles (1, √6/4, √6/4)	$\alpha/2$ = 35.26°	120°
Octahedron (O)	equilateral 1; 3 right isosceles (1, √2/2, √2/2)	$(\pi - \alpha)/2$ = 54.74°	90°
Dodecahedron (D)	regular pentagon 1; 5 isosceles	$\theta_D/2$ = 58.28°	120°
Icosahedron (Cleo-Icos)	equilateral 1; 3 isosceles, side/base √((5+√5)/8)	$\theta_I/2$ = 69.09°	72°

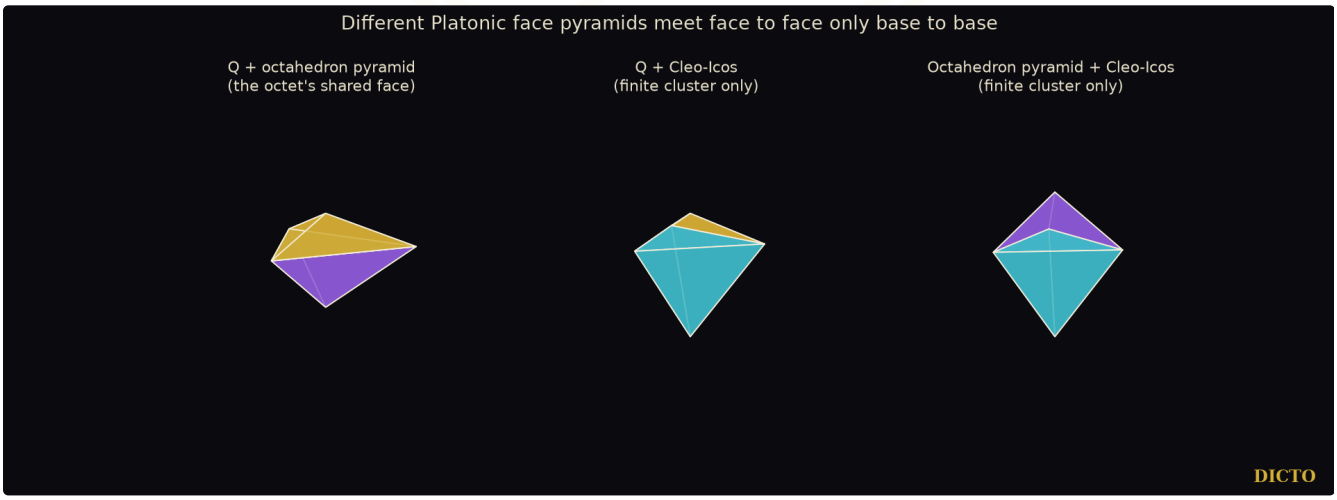


Figure 3.1. The three ways two different face pyramids meet face to face, all base to base: the tetrahedron's pyramid on the octahedron's (the octet's shared face), and the Cleo-Icos on either of them (finite only).

2. Shared networks, face to face

Theorem 1. In a face-to-face tiling by face pyramids, two different face pyramids meet only base to base, and the only tiling that uses two different face pyramids is made of the tetrahedron's and the octahedron's pyramids: the octet truss cut from its centres, and its layer stackings. The Cleo-Icos meets another face pyramid only in finite bipyramids and never in a face-to-face space filling.

Proof. (1) The side faces of different pyramids are never similar: the cosines of their apex angles, 1/3, –1/3, 0, √5/3 and 1/√5, all differ, and no side face is equilateral; so a side face meets only a side face of the same pyramid, and congruence then carries base edge to base edge and apex to apex. (2) The square and the pentagon occur once each; the equilateral bases of the tetrahedron's, octahedron's and icosahedron's pyramids match each other at equal edge. So two different pyramids meet only base to base. (3) Round a side edge every neighbour is the same kind with the same apex, so the 360° is made by 3, 3, 4, 3 or 5 copies, and each pyramid lies inside its whole solid. Round a base edge the pyramids come in pairs of one kind glued on their sides, and a mixed edge star needs as many angles α/2 as (π – α)/2, since these angles are independent over the rationals of π (Chapter 2, Lemma 2): the only mixed star is four of each, 4·α/2 + 4·(π – α)/2 = 2π, the octet edge. Every other mixed star allowed by the angles alone (twenty are listed in the study data) fails the face condition. ∎

So the pyramids are no more sociable than their solids. A network built on the Cleo-Icos needs a partner that is not a face pyramid: Antonius, which fills the golden rhombohedron with two Cleo-Icos (#20, Chapter 2), or the connectors of #47, with which all five Platonic solids, and so all five face pyramids, fill space together.

3. New deltahedra from DICTO's pieces?

Seventeen of DICTO's pieces and their relatives carry at least one equilateral face: the regular tetrahedron, octahedron and icosahedron, the half octahedron P, Antonius, the Admiral Cap, the six Mosseri–Sadoc tiles (among them AXE and CLEO, Chapter 1, and the Tricap), the Weirdstone and the Tadpole, and the three triangle-based face pyramids. We glued them whole face to whole congruent face, in every way, mirror images allowed, keeping only clusters whose interiors do not meet, and kept a result when its surface, merged into planar faces, was a closed surface of equilateral triangles of one size enclosing the pieces' total volume. Run A used all pieces and is complete to three pieces (23, 262, 7,162 and 120,020 clusters at one to four pieces); run B used DICTO's pieces alone and reached five (363,319 clusters).

Result. 772 different solids with equilateral faces were found, every one closed. Four have the face pattern of a convex deltahedron and are classical: four Q make the tetrahedron, two P the octahedron, two tetrahedra the triangular bipyramid, one icosahedron itself. None is a variant of a convex deltahedron, with its pattern of faces but another shape. The other 768 are face-gluings of regular tetrahedra, octahedra and icosahedra, a known infinite class; the smallest lie inside the enumeration of deltahedral graphs up to ten corners by Tsuruta, Mitani, Kanamori and Fukui (2014). No solid uses Antonius, the Admiral Cap, AXE, CLEO, the Tricap, the Weirdstone, the Tadpole, the Cleo-Icos or the octahedron's pyramid. Six Admiral Caps on an icosahedron give a deltahedron only all together, as Pacioli's octahedron (#47).

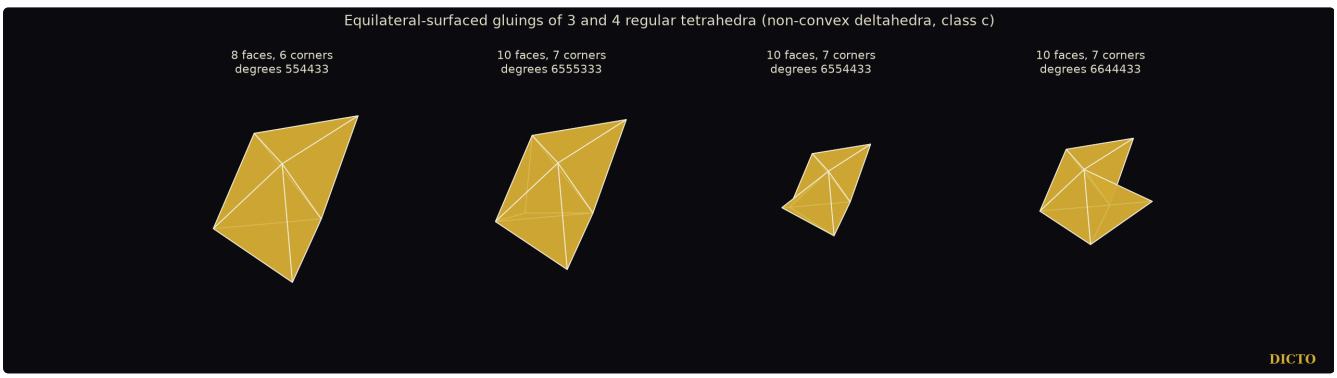


Figure 3.2. The smallest solids of the search that are not regular: face-gluings of three and of four regular tetrahedra.

Variants of the regular deltahedra, which DICTO expected to join differently, do exist: one face pattern can have a convex form and non-convex ones (Tsuruta et al., graph 9_{50}). They come from denting a solid, pushing a corner in, which removes material; gluing pieces never makes them in this range. Chapter 4 finds such variants where DICTO looked next: in the shadows and slices of the four-dimensional solids.

4. How the solids found can join

The counting map of Chapter 2 decides the rest. With $h(\alpha) = 1$, a solid of n_T regular tetrahedra, n_O octahedra and n_P half octahedra (edge 1) scores $6n_T - 12n_O - 6n_P$; with $h(\theta_I) = 1$, a solid with n_I icosahedra scores $30n_I$. Every solid with an icosahedron scores above zero, so it fills space neither alone nor with face pyramids. A solid of tetrahedra and octahedra with equilateral faces never has a tetrahedron on an octahedron face (the faces would lie in one plane), so it is all tetrahedra or all octahedra and its score is not zero: none of the 772 solids fills space alone.

5. Where the results stop

Gluings of five or more pieces with the regular solids, and of six or more of DICTO's pieces alone, were not searched; the search grows about fourteen times per piece. Space fillings of tetrahedron gluings with octahedron pyramids, and of icosahedra with Antonius, are not excluded by the counting map and were not decided. Face shapes, angles, face matching and the counting arguments are exact; the search, its overlap tests and the sorting into distinct solids are in floating point and are a screening.

6. Conclusion

Face to face, the five pyramids keep to their solids: only the tetrahedron's and the octahedron's share a network, the octet, and the Cleo-Icos finds partners only outside the family. Glued together, DICTO's pieces make no new deltahedron up to four pieces (five for his own pieces). The new shapes DICTO was looking for lie elsewhere: in the four-dimensional solids, whose shadows and slices are the subject of Part II, and whose cores, Part III shows, are exactly these five pyramids.

Sources checked

DICTO's reference library, five queries ("non-convex deltahedra enumeration"; "deltahedron equilateral triangle faces polyhedra list"; "face-to-face gluing of regular tetrahedra octahedra icosahedra polyhedra"; "central pyramids of Platonic solids space filling"; "Mosseri Sadoc golden tetrahedra tiles"), with the items returned and read: Johnson 1966; Sommerville 1922; Kramer 1982; Torquato and Jiao 2009; Lagarias and Zong 2012; Koca, Koc, Koca and Al-Siyabi 2020; Papadopolos and Ogievetsky 1999; Akpanya, Goertzen and Niemeyer 2024; and others listed in the study. Web: the search "non-convex deltahedra enumeration Cundy Trigg Freudenthal van der Waerden", with en.wikipedia.org/wiki/Deltahedron and Tsuruta et al. 2014 (ICGG 2014, pages 1–4) read. DICTO's papers #19, #20, #44 and #47.

Part II. Shadows



Chapter 4. Shadows of the Fourth Dimension

From paper #62, DICTO DOI 10.5281/zenodo.23687223

Deltahedra that join differently: the triangle-faced shadows and slices of the 5-, 16-, 24- and 600-cell

Abstract. A deltahedron is a solid whose faces are all equilateral triangles; only eight of them are convex. Four regular solids of four-dimensional space are also built from triangles: the 5-cell, the 16-cell, the 24-cell and the 600-cell. DICTO's idea was that their three-dimensional shadows and slices might stand closer to a new family of deltahedra — same pattern of triangles, different shape — than to the eight regular ones, and that such variants would join to other pieces differently. We compute every shadow and slice of the four solids along their four main directions exactly, in numbers of the form $a + b\sqrt{5}$, and find deltahedra of two new kinds. (a) Two shadows are variants of convex deltahedra, and each joins differently from its regular counterpart: the 5-cell's edge-first shadow (TBP-5) has the pattern of the triangular bipyramid J12, and three copies of it close round an edge; the 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron, and it fills space on its own. (b) Four shadows and slices of the 600-cell can be rebuilt exactly with every edge equal, keeping their pattern of triangles, and all four are non-convex: a 12-dented icosahedron; the excavated and the augmented dodecahedron; Shadow-17, two diminished icosahedra (J11) glued on their pentagons, identified through DICTO's idea of cutting the pattern in halves; and Slice-48, a solid of 92 triangles with tetrahedral symmetry, found by solving the 600-cell's six large slices exactly with their full symmetry. Supporting results: apart from copies of the solids' own cells and vertex figures, no shadow, layer or slice is a regular deltahedron (Proposition 1), and the other five large slices have no equilateral form with their full symmetry (§6).

1. The idea

DICTO's idea (27 October 2026) was that "4th dimensional shadows may more closely represent any new deltahedron type family than they do the original ones", and that such variants, set against the regular ones, would be "different shapes with different attachability". On the same day a search for new deltahedra glued from DICTO's pieces (AXE, CLEO, Cleo-Icos, Admiral Cap, Tricap and others; up to 4 pieces of all kinds, up to 5 of DICTO's) had found 772 solids with all faces equilateral, every one of them a gluing of regular tetrahedra, octahedra and icosahedra: no new deltahedron and no variant. The fourth dimension was the next place to look, and it gave the following.

Main result. The triangle-faced shadows and slices of the 5-, 16-, 24- and 600-cell give deltahedra of two new kinds.
(a) **Variants that join differently** (§4). The 5-cell's edge-first shadow (TBP-5) has the pattern of the triangular bipyramid J12 with unequal edges, and three copies of it close face to face round an edge. The 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron with unequal edges, and it fills space on its own.
(b) **Equal-edged solids from the 600-cell** (§5, §6). Four of the 600-cell's shadows and slices can be rebuilt exactly with every edge the same length, keeping their pattern of triangles: the vertex-first shadow (42 vertices, 80 triangles) as a 12-dented icosahedron; the slices at the dodecahedron height (32 vertices, 60 triangles) as the excavated and the augmented dodecahedron; the outer edge-first slices (17 vertices, 30 triangles) as Shadow-17, two J11 glued on their pentagons; and the 48-vertex cell-first slice (92 triangles) as Slice-48, with tetrahedral symmetry. All four forms are non-convex.

2. Method

All arithmetic is exact in $\mathbb{Q}(\sqrt{5})$ (numbers $a + b\sqrt{5}$, a and b rational, with exact signs); floating point only proposes convex hulls, and every hull face is then checked exactly (all points on one side, every edge in two faces, Euler characteristic 2). For each solid and each of four directions — through a vertex, the middle of an edge, the centre of a triangle and the centre of a cell — we take the **shadow** (the outline of the projection into the 3-space at right angles), the **layers** (the outline of the vertices at one height) and the **slices** (the true cut by a 3-space at right angles, through the centre and at every vertex height: the vertices on it plus every edge crossing it). This gives 223 records and 157 distinct solids, 27 of which have all faces triangles. Each of the 27 is compared with the eight convex deltahedra by graph isomorphism; for the patterns that match none of the eight, an equilateral form is sought (least squares on the edge lengths, as in Tsuruta et al. 2014) and confirmed exactly where found.

The shadows and slices themselves are convex, as every projection and cut of a convex solid is, and in general their edges are unequal. The question for each is whether its *pattern* of triangles can be rebuilt with all edges equal. Where that needs some edges folded inwards, the equilateral form is non-convex: the dents belong to that form, not to the 4D solid.

3. Results

The table lists the all-triangle shadows and slices other than regular tetrahedra, octahedra and icosahedra, grouped by pattern, with the equilateral form found for each pattern.

4D source	Solid	Pattern	Edge ratio	Equilateral form
5-cell	edge-first shadow (TBP-5)	triangular bipyramid J12	$\sqrt{(12/7)}$ = 1.309	variant of J12
16-cell	edge-first shadow = central slice (OCT-16)	octahedron	$2/\sqrt{3}$ = 1.155	variant of the octahedron
600-cell	vertex-first shadow = central slice (42, 80)	pentakis icosidodecahedron	$2/\sqrt{3}$	12-dented icosahedron (exact)
600-cell	vertex-first slices at the dodecahedron height (32, 60)	pentakis dodecahedron	1.144	augmented dodecahedron; excavated dodecahedron (both exact)
600-cell	edge-first slices (17, 30): Shadow-17	none of the 8	2.927	two J11 on their pentagons (exact)
16-, 600-cell	face- and edge-first (8 or 12 vertices)	hexagonal, decagonal bipyramids	1.41–2.29	none can exist (Prop. 2)
600-cell	six large slices (28–72 vertices)	none of the 8	2.29–4.81	§6

All other all-triangle results are the regular tetrahedron, octahedron or icosahedron, and each is a cell or vertex figure of its 4D solid. Two supporting propositions fix what the table leaves out.

> **Proposition 1.** No shadow, layer or slice of the 5-, 16-, 24- or 600-cell along the four main directions is a regular deltahedron, except copies of the solid's own cells and vertex figures. Among the others, the only patterns of a convex deltahedron that occur are those of the triangular bipyramid and the octahedron, each with unequal edges.

> **Proposition 2.** A bipyramid over an n-gon with all edges equal exists only for $n \le 5$: for $n \ge 6$ the equilateral apex would lie in the plane of the polygon or the solid would cross itself. So the hexagonal and decagonal bipyramid patterns of the 16- and 600-cell have no equilateral form.

4. Variants join differently

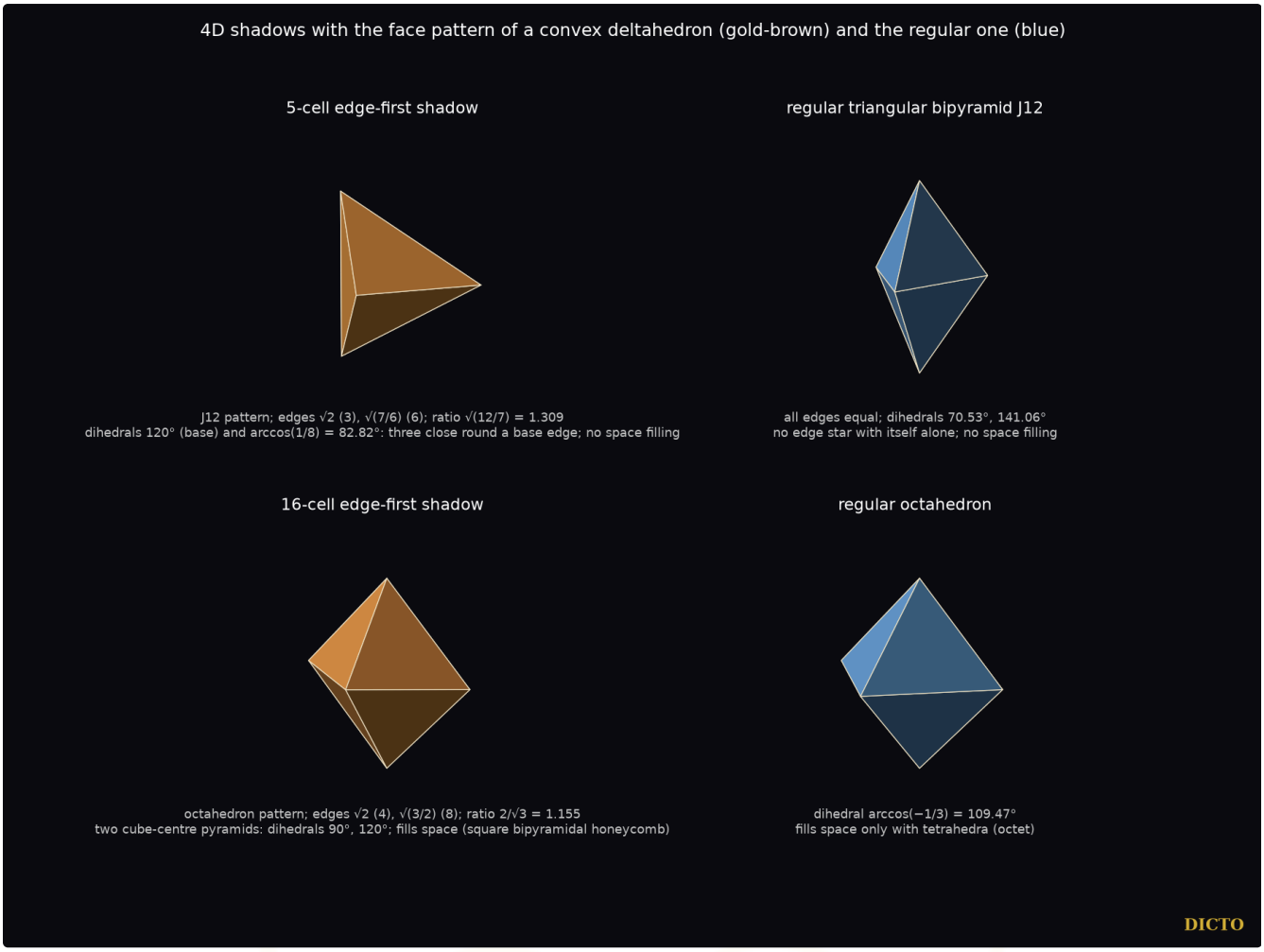


Figure 4.1. The two variants (gold-brown) beside their regular counterparts (blue). Top: the 5-cell's edge-first shadow, with the pattern of the triangular bipyramid J12. Bottom: the 16-cell's edge-first shadow, with the pattern of the octahedron.

TBP-5, the 5-cell's edge-first shadow, has three edges of length √2 and six of length √(7/6). Its faces match no face of J12. Its base edges have a dihedral angle of exactly 120°, so three copies of TBP-5 close face to face round a base edge; regular copies of J12 close no such ring. Its side edges have dihedral angle arccos(1/8), which closes no star of copies, so neither TBP-5 nor J12 fills space.

OCT-16, the 16-cell's edge-first shadow, is a square bipyramid: two copies of the pyramid that joins the centre of a cube to one of its faces, glued base to base. Its dihedral angles are 90° and 120°, and it fills space on its own, face to face: it is the cell of the square bipyramidal honeycomb, a known tiling (§8). The regular octahedron fills space only together with tetrahedra. The angles alone would allow a mixed star round an edge (2·90° + 70.53° + 109.47° = 360°), but the faces of OCT-16 do not match those of the regular octahedron and tetrahedron, so the variant and the regular family do not join face to face.

Both variants therefore differ from their regular counterparts in shape and in how they join, as DICTO expected.

5. The 600-cell's dented patterns, and Shadow-17

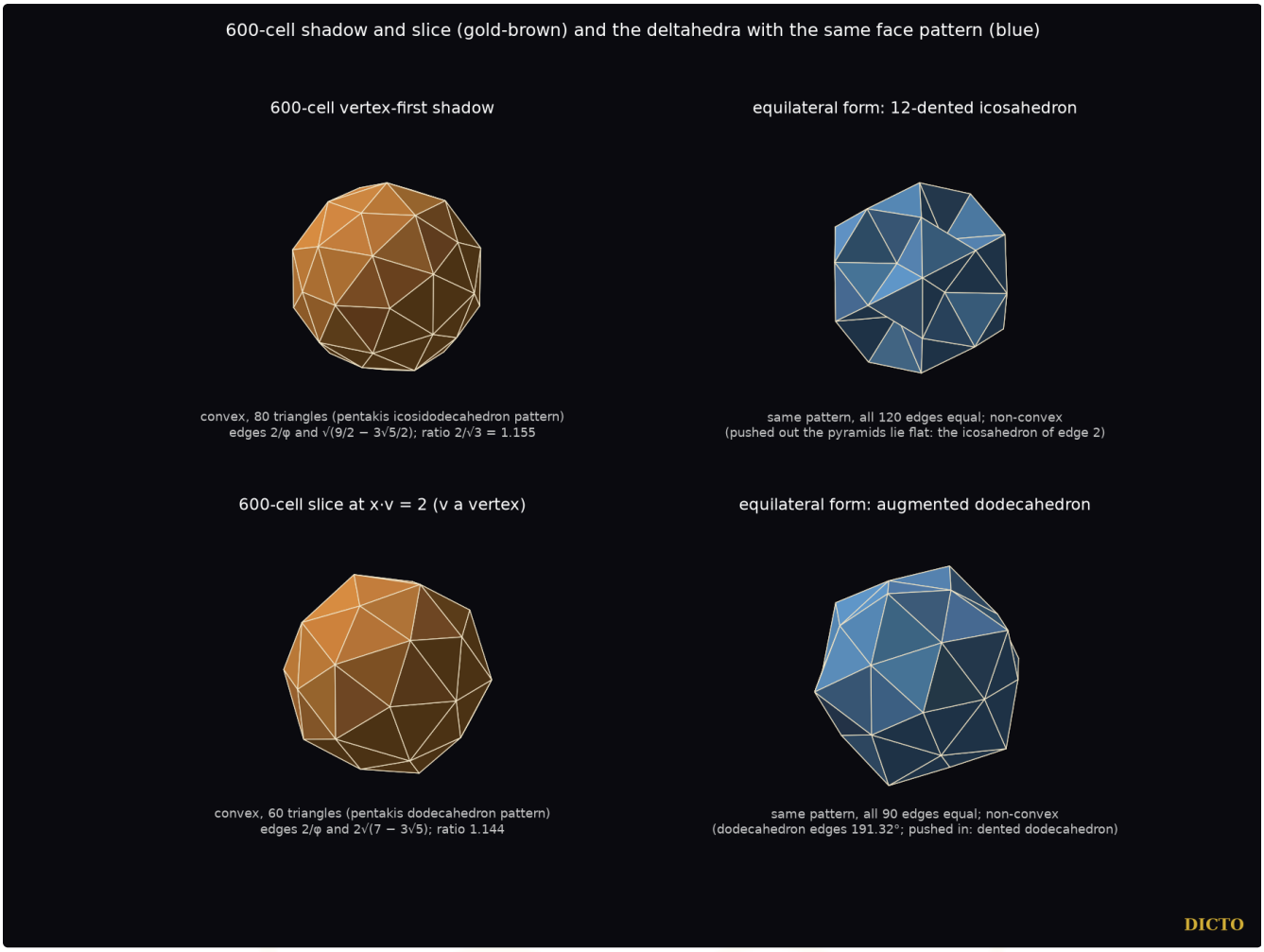


Figure 4.2. The 600-cell's vertex-first slices: the patterns of the pentakis icosidodecahedron and the pentakis dodecahedron, and their exact equilateral forms (12-dented icosahedron; augmented and excavated dodecahedra).

The 600-cell's vertex-first shadow is also its central slice, with 42 vertices and 80 triangles. Its equilateral form is an icosahedron with a shallow dent at each vertex (placeholder name: 12-dented icosahedron). The vertex-first slices at the dodecahedron height (32 vertices, 60 triangles) have two equilateral forms: the excavated dodecahedron, a known deltahedron, with its pyramids dented inward, and the augmented dodecahedron, with its pyramids pointing outward. The names 12-dented icosahedron and augmented dodecahedron are placeholders; both describe obvious constructions, and neither solid is claimed new.

Shadow-17. The 600-cell's edge-first slices at the outer heights have 17 vertices and 30 triangles and five-fold symmetry: a tip at each end and three rings of five vertices between them, with the vertices of the middle ring of degree 6. An equilateral form was first found numerically. DICTO's idea of cutting the pattern in halves identified it: cut at the middle ring, and each half is a J11, the gyroelongated pentagonal pyramid (the diminished icosahedron). **Shadow-17's equilateral form is two J11 glued on their pentagons.** This form is exact by construction, with every edge 1, and all 136 corner-to-corner distances of the numerical form agree with it to $4{\cdot}10^{-16}$. Its five folded edges are the edges of the shared pentagon, where the dihedral angles of the two J11 add to $2 \times 100.81° = 201.62°$. J11 is the piece with which DICTO's work began: it fills the gaps between dodecahedra and icosahedra.

6. Slice-48 and the six large slices

One of the 600-cell's six larger all-triangle slice patterns, the 48-vertex cell-first slice, has an exact equilateral form with its full symmetry: Slice-48, described at the end of this section. This section gives the method by which

all six were solved and the results for each.

The six patterns have 28 to 72 vertices, with rings of high-degree vertices (7 to 10 triangles meeting at a vertex) between tips of degree 5. Cutting them in halves or caps, as for Shadow-17, matched no known piece, so each was solved directly. The exact symmetry group of each slice equals the full symmetry group of its triangle pattern. Keeping that symmetry, one vertex of each orbit is placed in the subspace fixed by its stabiliser, and each orbit of edges gives one equation $|\text{edge}|^2 = 1$; every system is square (as many equations as unknowns). The number of complex solutions of each system was found exactly over $\mathbb{Q}(\sqrt{5})$ (Singular); every solution was found and refined to 120 digits, and a system counts as complete only when the number of solutions found equals the exact count. Every real solution was then tested for a proper embedding.

Slice (600-cell)	Vertices, triangles	Symmetry	Solutions (complex / real)	Equilateral form
vertex-first, at ± $(\sqrt{5}-1)$	72, 140	Ih	8 / 4	none: vertices coincide
edge-first, at ± $(3+\sqrt{5})/2$	32, 60	D5d	128 / 48	none: edges pierce faces, or faces touch
face-first, at ± $(4+2\sqrt{5})/3$	30, 56	D3d	464 / 104	none: edges pierce faces, or faces touch
cell-first, at ± $(5+3\sqrt{5})/4$	28, 52	Td	32 / 12	none: edges pierce or cross, vertices coincide
cell-first, at ± $(3+\sqrt{5})/2$	42, 80	Td	96 / 32	none: every solution crosses itself
cell-first, at ± $(2+\sqrt{5})/2$	**48, 92**	**Td**	**192 / 48**	**Slice-48** (one shape; its two solutions differ by a half-turn)

The systems of the 72-, 32-, 28- and 42-vertex slices were also solved with only their rotation group kept, and again no form exists; the real solution counts of the same four systems were rechecked exactly.

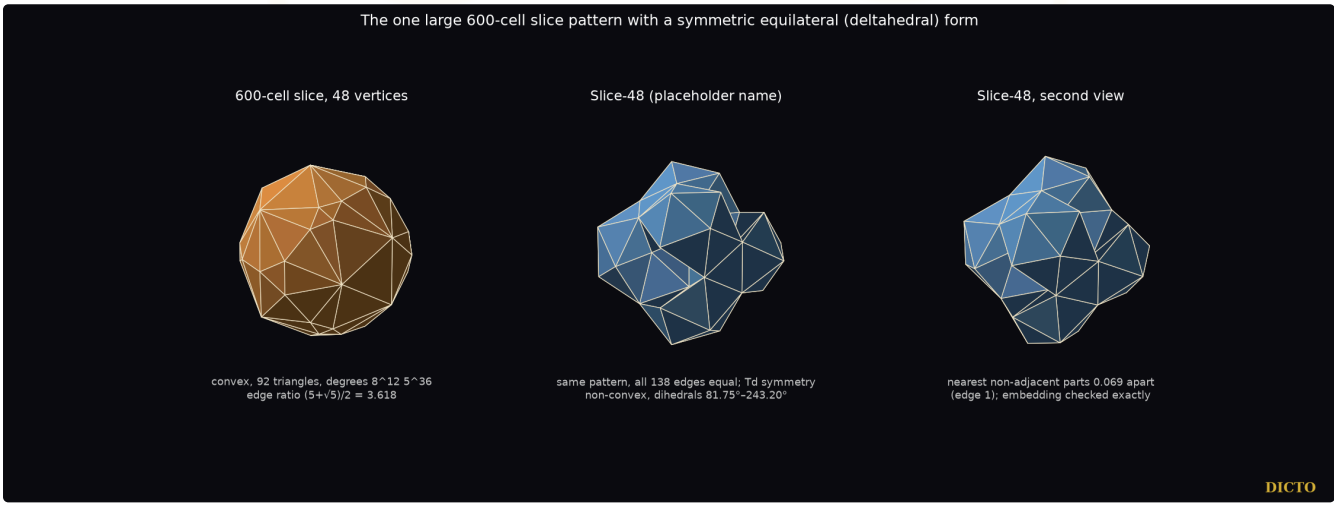


Figure 4.3. The 600-cell's 48-vertex cell-first slice (left, convex, edges in ratio up to $(5+\sqrt{5})/2$) and Slice-48, its equilateral form, from two views.

> **Slice-48** (placeholder name). The 48-vertex cell-first slice of the 600-cell has exactly one equilateral form with its tetrahedral symmetry Td: 48 vertices (12 of degree 8, 36 of degree 5), 138 equal edges and 92 triangles. It is not convex: 24 of its vertices lie on its hull, and its dihedral angles are 81.75°, 130.98°, 135.83°, 143.83°, 191.03°, 197.94°, 216.17° and 243.20°. At edge length 1, its nearest non-adjacent parts are 0.069 apart; the embedding was confirmed with exact orientation tests on 85-digit rational coordinates.

Unlike Shadow-17, Slice-48 contains no piece of the icosahedron: it has no dihedral angle of 138.19° and no pentagonal cap. It is not among the non-convex deltahedra we checked (§8); since we did not read all the

original papers, we do not claim it new. Forms with less symmetry than the slice's group are not excluded; the rotation-only systems of the 30- and 48-vertex slices did not finish in 50 minutes.

7. A note on the 600-cell's sections

The vertex-first "sections" of the 600-cell that are often listed — icosahedron, dodecahedron, icosahedron, icosidodecahedron — are the outlines of its vertex layers. A true cut at the dodecahedron height or at the equator also meets edges that cross it, and adds their midpoints: the true central section is the 42-vertex solid of §5, with the same pattern, edge lengths and angles as the shadow. This was checked exactly and with an independent floating-point computation.

8. Prior art and credit

Coxeter's *Regular Polytopes* treats the polytopes, their projections and the vertex-first sections of the 600-cell. The eight convex deltahedra are classical (Freudenthal and van der Waerden 1947); non-convex equilateral realisations and dented variants of deltahedral patterns are studied by Tsuruta et al. (2014). For Slice-48 we also checked the biform deltahedra (Cundy; Olshevsky), the isohedral deltahedra (Shephard; McNeill) and Trigg's spiral deltahedra, through their Wikipedia summaries. The square bipyramidal honeycomb and the excavated dodecahedron are known. In the sources checked, we did not find the comparison of the 4D shadows and slices with the deltahedra, the attachability of the variants, or the identification of Shadow-17.

9. Conclusion

> **Main result.** The triangle-faced shadows and slices of the four-dimensional solids give deltahedra of two new kinds.
> (a) **Variants that join differently.** The 5-cell's edge-first shadow (TBP-5) has the pattern of J12, and three copies of it close round an edge. The 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron, and it fills space on its own.
> (b) **Equal-edged solids from the 600-cell.** Four of its shadows and slices can be rebuilt exactly with every edge the same length, keeping their pattern of triangles; all four forms are non-convex.

600-cell shadow or slice	Triangles	Rebuilt with equal edges
vertex-first shadow = central slice (42 vertices)	80	12-dented icosahedron: an icosahedron with a shallow dent at each of its 12 vertices
vertex-first slices at the dodecahedron height (32 vertices)	60	excavated dodecahedron (dents inward; known) and augmented dodecahedron (pyramids outward)
edge-first slices at the outer heights (17 vertices)	30	Shadow-17: two J11 glued on their pentagons
cell-first slices with 48 vertices	92	Slice-48: tetrahedral symmetry, dihedral angles 81.75° to 243.20°

Shadow-17 is built from two copies of J11, the piece with which DICTO's work began. Slice-48 is the largest of the four: 92 equal triangles with the full tetrahedral symmetry of its slice, found by solving all six large slices of the 600-cell exactly.

Part III. The Queens



Chapter 5. The White Queen 600-cell

From paper #63, DICTO DOI 10.5281/zenodo.23690856

An exact atlas of the slices and shadows of the 600-cell

Abstract. The 600-cell is the four-dimensional solid built from 600 regular tetrahedra, twenty round every vertex. DICTO proposed a definitive, proved picture of it. We give a complete exact atlas of its three-dimensional slices in its four main directions (through a vertex, the middle of an edge, the centre of a triangle and the centre of a cell). Every slice is computed exactly, in numbers of the form $a + b\sqrt{5}$ with a and b rational. Between two neighbouring vertex heights a slice keeps one pattern of faces, so the cuts at the vertex heights and one cut inside each gap give every slice pattern there is: 62 patterns on one side of the centre (9, 17, 21 and 15 in the four directions). Exactly twelve are made only of triangles, and we identify each one. The largest slices have 144 vertices. With the four shadows of the 600-cell, this completes the atlas. Each shadow resolves exactly into two halves, its front cells and its back cells, each filling it once; along the six turnings between the main views, the set of cells in front changes at only 14 moments. Round one vertex the 600 cells form 15 shells, each of one kind of piece, the innermost a core of 20 Cleo-Icos. The slices as they are, convex and exact, with faces of many sizes, are the subject of this chapter. A short side note records what happens when a slice is forced to have equal edges: of six large all-triangle slices only one can be. That slice is Humpty Dumpty, a convex egg of 92 triangles; its forced equal-edged form, Fallen Humpty, has lost the egg's rounded shape.

1. Begin at the beginning

> *"Begin at the beginning," the King said gravely, "and go on till you come to the end: then stop." — Alice's Adventures in Wonderland, ch. 12*

That is the plan of this chapter: start at the centre of the 600-cell and cut it, height by height, until the cut leaves it at the tip. DICTO proposed that the 600-cell could be assembled into a definitive picture, every part proved. Our previous paper (#62, Shadows of the Fourth Dimension) found which of its slices have only triangular faces, and which of those can be rebuilt with every edge equal; here we map every slice. The problem is old. Aristotle wrote that among solids the pyramid fills space (*On the Heavens* III.8); his commentators argued over how many tetrahedra meet round a point, twelve or twenty, and Thomas Bradwardine saw that the icosahedron cuts from its centre into twenty equal tetrahedra without deciding whether they are regular (Lagarias and Zong, 2012; our #44, Aristotle's Pyramid). They are not: they are the Cleo-Icos. Kepler nested the Platonic solids in shells (*Mysterium Cosmographicum*, 1596). In the 600-cell, found by Schläfli about 1852, twenty regular tetrahedra do close round every vertex, and §6 shows its 600 cells in shells round such a core. The title is DICTO's: the 600-cell belongs to the White Queen of *Through the Looking-Glass*, who practised believing impossible things (§7).

2. Method

The 600-cell is taken with its 120 vertices at distance 2 from the centre: the 8 permutations of (±2, 0, 0, 0), the 16 points (±1, ±1, ±1, ±1) and the 96 even permutations of (±φ, ±1, ±1/φ, 0), where $\varphi = (1+\sqrt{5})/2$. Its edges have length $\sqrt{5} - 1$. All arithmetic is exact in $\mathbb{Q}(\sqrt{5})$. For each direction d, the height of a point x is $x \cdot d$, and we sort the vertices by height. The slice at height h is the convex hull of the vertices at that height together with the points where edges of the 600-cell cross that height. Hulls are proposed in floating point, and every face is then verified exactly. The checks, run on every slice, are these. Repeated points are merged by exact comparison. For each proposed face, the plane through three of its corners is checked to support the slice: every point lies on it or strictly on one side, decided by the exact sign of a number in $\mathbb{Q}(\sqrt{5})$. The face is then all the points on that plane, ordered by an exact two-dimensional hull, and turned to face outward by an exact sign. Finally every edge of the

slice must lie in exactly two faces, and the counts must satisfy V − E + F = 2. A closed surface of supporting faces is the boundary of the hull, so a face missed in floating point would show as an edge in only one face.

> **Lemma (one pattern per gap).** Let $h_1 < h_2$ be two neighbouring vertex heights. At every height strictly between them the slice has the same pattern of faces. *Proof.* An edge of the 600-cell crosses the cut at height h exactly when its two ends lie on opposite sides of h. No vertex has a height strictly between h_1 and h_2, so as h moves through the gap no vertex changes side, and the same edges cross the cut. Inside the gap each face of the slice is the cross-section of one tetrahedral cell, with its corners at the crossing points on that cell's edges, and two faces share an edge of the slice exactly when their cells share a triangle that is cut. Which cells and triangles are cut, and through which edges, is therefore the same throughout the gap, and so is the pattern of faces. Hence the cuts at the vertex heights and one cut inside each gap give every slice pattern. The lemma is about the pattern only: edge lengths and angles change with the height.

In each direction, d is the vertex, the edge midpoint, the triangle centre or the cell centre itself, not a unit vector: vertex-first d = (2, 0, 0, 0); edge-first d = ((5+√5)/4, 1/2, (√5−1)/4, 0), the midpoint of the edge from (2, 0, 0, 0) to (φ, 1, 1/φ, 0); face-first d = (1+√5/3, 2/3, 0, 0), the centre of the triangle with corners (2, 0, 0, 0), (φ, 1, 1/φ, 0) and (φ, 1, −1/φ, 0); cell-first d = ((7+3√5)/8, (3+√5)/8, 0, 1/4), the centre of the cell with corners (2, 0, 0, 0), (φ, 1, 1/φ, 0), (φ, 1, −1/φ, 0) and (φ, 1/φ, 0, 1). Each d is the average of the corners named. Since d is not a unit vector, heights compare within one direction only. The 600-cell is centrally symmetric, so the far half of each direction mirrors the near half; we list the near half, from the centre to the tip.

3. The atlas

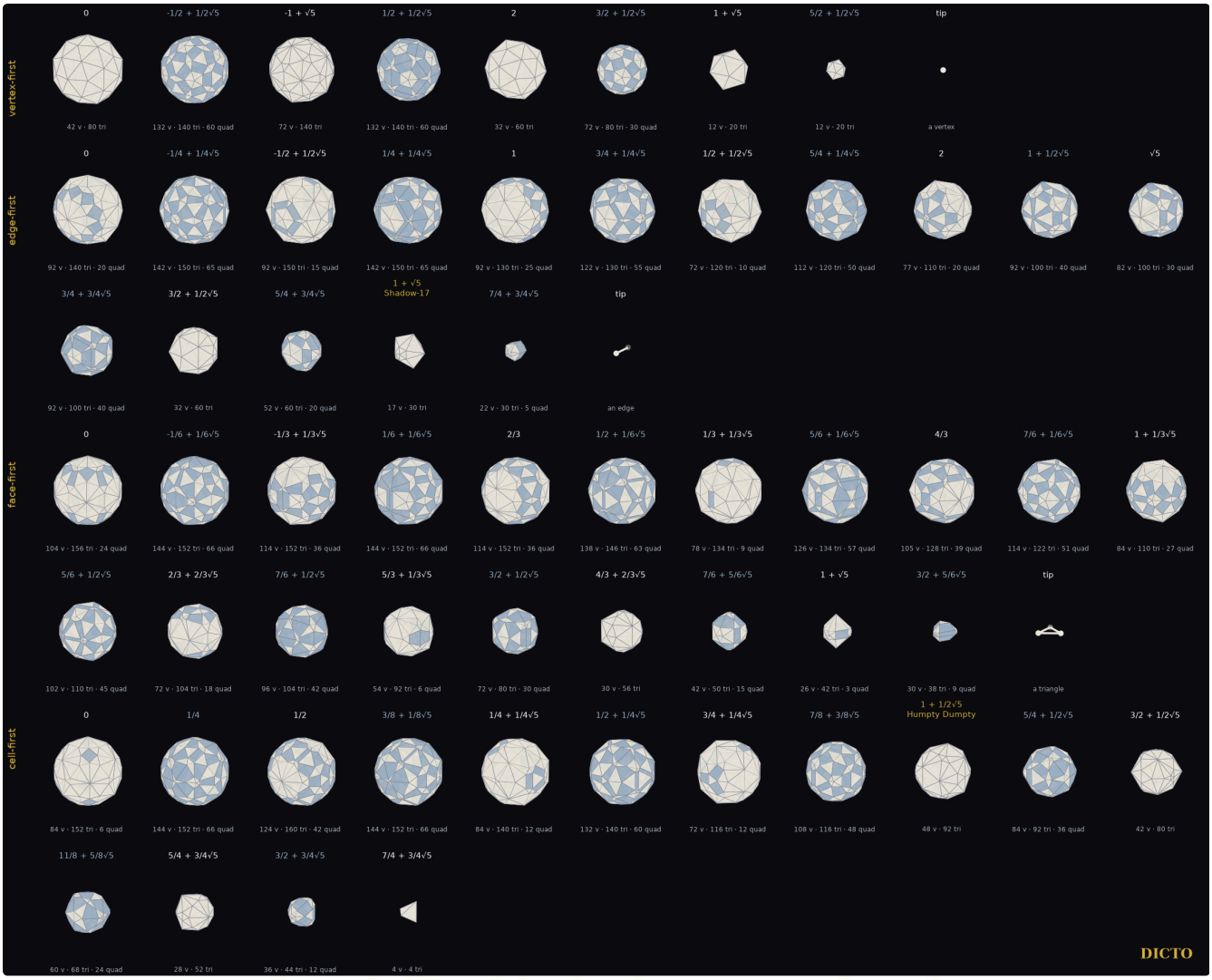


Figure 5.1. Every slice pattern of the 600-cell, from the centre to the tip, read left to right and row by row, in the four directions; one scale throughout. White labels: slices at a vertex height; blue labels: the pattern that holds over the whole gap to the next height. Triangles ivory, four-sided faces blue. Gold: Shadow-17 (#62) and Humpty Dumpty (§7).

Direction	Vertex heights (centre to tip)	Gaps	Slice patterns	Tip	Largest slice
vertex-first	5	4	9	a vertex	132 vertices
edge-first	9	8	17	an edge	142 vertices
face-first	11	10	21	a triangle	144 vertices
cell-first	8	7	15	a tetrahedron	144 vertices
total			**62**		

Every slice has only triangles and four-sided faces. Each face is the cross-section of one tetrahedral cell, a triangle or a four-sided face, or else a triangle of the 600-cell lying in the cut. A four-sided face appears exactly where the cut crosses a cell through four of its edges. The atlas, which cuts each gap at its middle height, has 1636 four-sided faces, and exactly 46 of them are squares, checked exactly. The squares are exactly the cuts through the middle of a cell, parallel to two opposite edges: each is the equator of the regular octahedron that joins the midpoints of that cell's six edges (edge $(\sqrt{5}-1)/2 = 1/\varphi$). Moved off the middle of the cell, such a cut is a rectangle. The 104 kites are explained in the same way: a kite appears exactly where a cell sits symmetrically across the cut, turned so that its four corners lie at four different heights, $-p$, $-q$, q and p from the cut. The half-turn of the cell that exchanges the corners at $-p$ and p, and those at $-q$ and q, reverses the heights and so maps the cut onto itself; on the four-sided face it fixes two opposite corners and exchanges the other two, a mirror

along a diagonal (checked exactly: 104 kites, 104 such cells). The remaining faces are rectangles, isosceles trapezia and general quadrilaterals.

4. The twelve all-triangle slices

> **Theorem.** Exactly twelve of the 62 slice patterns are made only of triangles: eleven at vertex heights and one inside a gap.

Direction	Height	Vertices, triangles	What it is
vertex-first	0 (centre)	42, 80	the central slice; equal-edged form: the 12-dented icosahedron (#62)
vertex-first	$\sqrt{5} - 1$	72, 140	no equal-edged form with its full symmetry (#62)
vertex-first	2	32, 60	equal-edged forms: excavated and augmented dodecahedron (#62)
vertex-first	$1 + \sqrt{5}$	12, 20	a regular icosahedron: the twelve neighbours of the tip vertex
vertex-first	the gap from $1 + \sqrt{5}$ to 4	12, 20	a regular icosahedron: the vertex figure, cut across the twelve edges that meet the tip
edge-first	$(3+\sqrt{5})/2$	32, 60	no equal-edged form with its full symmetry (#62)
edge-first	$1 + \sqrt{5}$	17, 30	Shadow-17: two J11 glued on their pentagons (#62)
face-first	$(4+2\sqrt{5})/3$	30, 56	no equal-edged form with its full symmetry (#62)
cell-first	$1 + \sqrt{5}/2$	48, 92	**Humpty Dumpty**, a convex egg; forced to equal edges it becomes Fallen Humpty (§7)
cell-first	$(3+\sqrt{5})/2$	42, 80	no equal-edged form with its full symmetry (#62)
cell-first	$(5+3\sqrt{5})/4$	28, 52	no equal-edged form with its full symmetry (#62)
cell-first	$(7+3\sqrt{5})/4$ (tip)	4, 4	a regular tetrahedron: the cell at the tip

The other 50 slice patterns all contain four-sided faces. Of the twelve, the two regular icosahedra and the tetrahedron are the 600-cell's own vertex figure and cell; the other nine are the all-triangle patterns studied in #62.

5. The four shadows

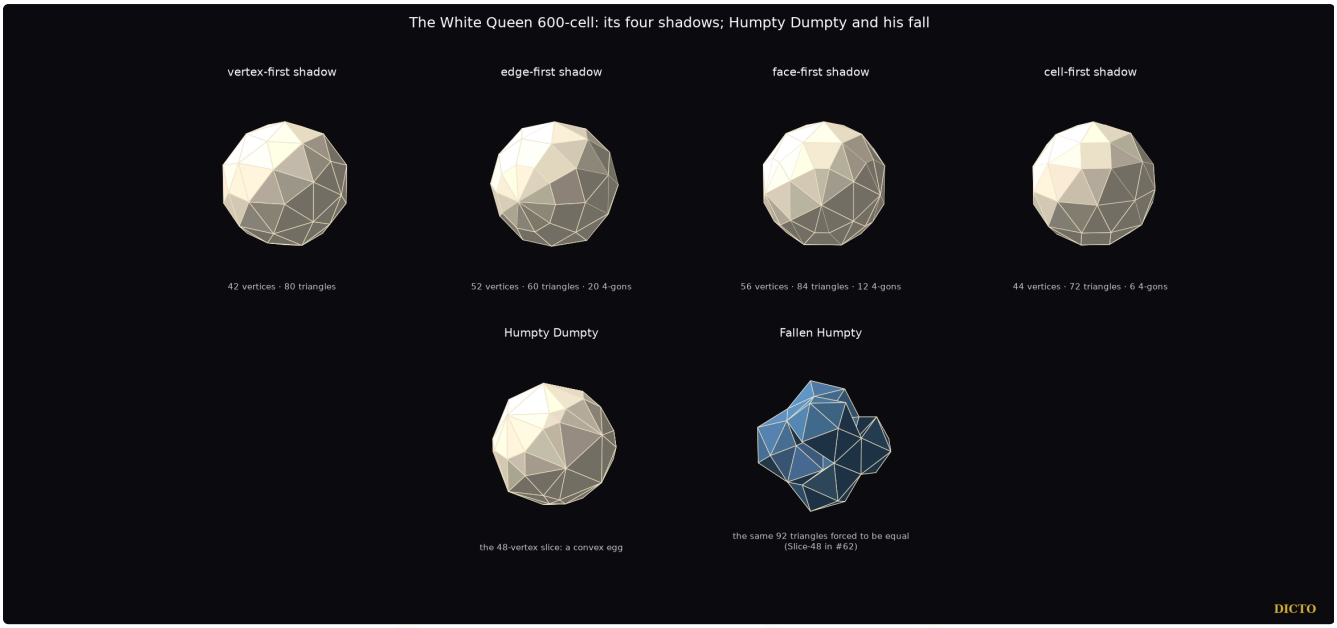


Figure 5.2. Top: the four shadows of the 600-cell. Bottom: Humpty Dumpty, the 48-vertex cell-first slice, a convex egg with edges of several lengths; and Fallen Humpty, the same 92 triangles forced to be equal (§7).

The shadows are the projections of the 600-cell along the four directions, and all four are convex. The vertex-first shadow has 42 vertices and 80 triangles, the same pattern as the central slice (#62). The edge-first shadow has 52 vertices, 60 triangles and 20 four-sided faces; the face-first shadow 56 vertices, 84 triangles and 12 four-sided faces; the cell-first shadow 44 vertices, 72 triangles and 6 four-sided faces.

6. The shadows resolved, and the Cleo shells

Halves. In each main view, every cell of the 600-cell faces the viewer, faces away, or is seen side-on, decided exactly by the sign of (cell centre)·d. The front cells fill the shadow exactly once, without gaps or overlaps, and so do the back cells. Checked exactly: in each view the projected volumes of the front cells add up to the volume of the shadow, and so do those of the back cells. The side-on cells flatten into the shadow's surface. The front cells together are the shadow's front layer. Two front cells are of one kind when their projections have the same six edge lengths.

View	Front	Back	Side-on	Kinds of front piece
vertex-first	270	270	60	7
edge-first	290	290	20	22
face-first	288	288	24	33
cell-first	273	273	54	20

The six turnings. Any two of the four main views span a plane in which the view can turn from one to the other: six turnings in all. Along a turning the front layer changes only at a moment when a cell turns side-on. At such a moment cells change sides in opposite pairs, a cell coming forward as the cell opposite it goes back, so the number in front stays the same. Between two such moments the front layer keeps one pattern (the shadow's counterpart of the lemma of §2). Computed exactly in $\mathbb{Q}(\sqrt{5})$, the six turnings (vertex–edge 18°, vertex–triangle 20.905°, vertex–cell 22.239°, edge–triangle 10.812°, edge–cell 13.283°, triangle–cell 7.761°) contain 14 such moments and 20 patterns between them; with the four main views, this accounts for every view along the turnings. Away from the moments the front layer holds 300 cells, or 298 or 294 where 4 or 12 cells stay side-on along the whole turning. One moment on the vertex–cell turning lies 15.52° from the vertex-first view, at the angle whose cosine is $(1 + 3\sqrt{5})/8$: that is 180° minus the 600-cell's dihedral angle, 164.48°.

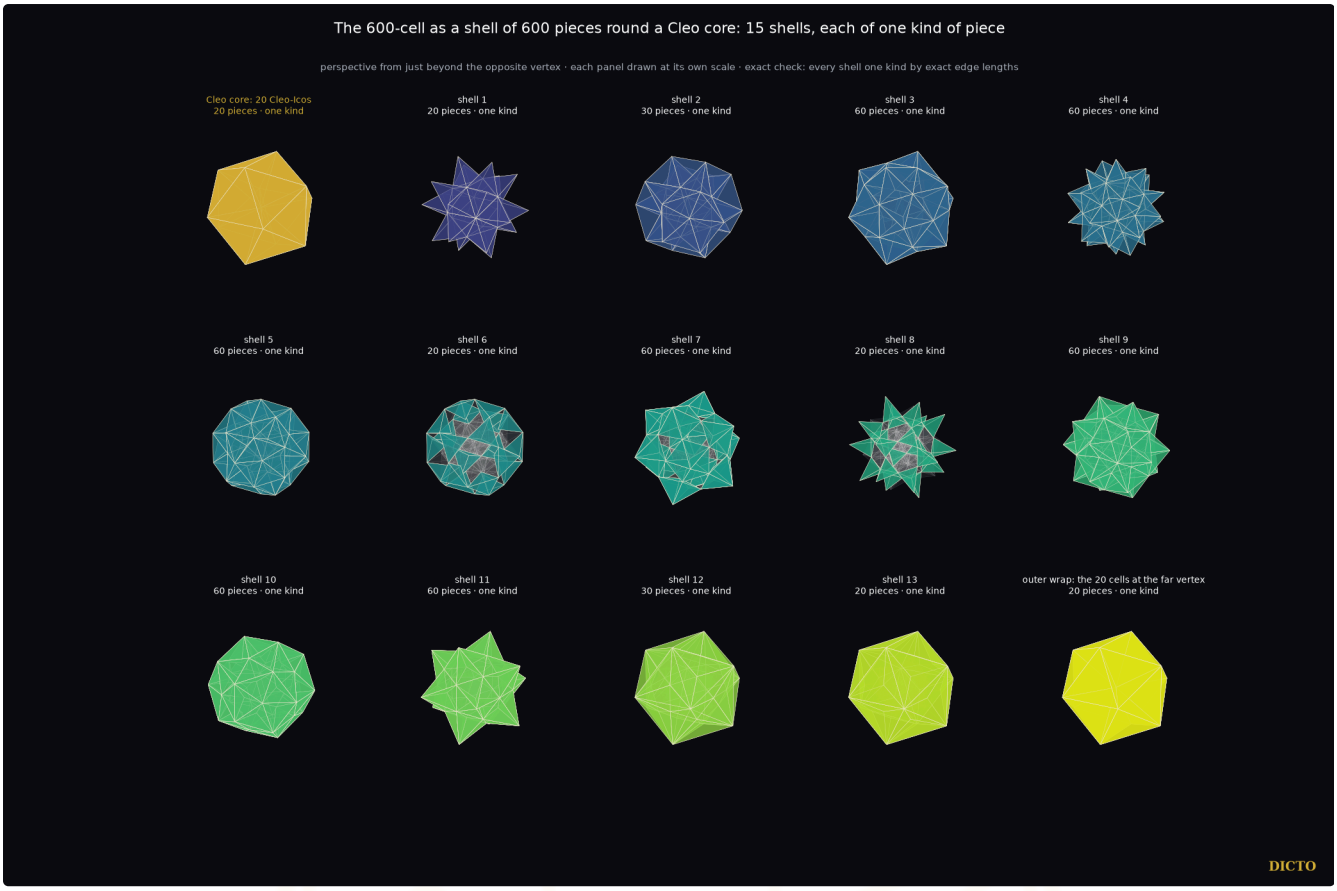


Figure 5.3. The 600-cell's 600 cells in 15 shells round one vertex, seen in perspective from just beyond the opposite vertex; each panel shows one shell solid and the shells inside it faint, at its own scale. Gold: the core of 20 Cleo-Icos.

The Cleo shells. Grouped by the height of their centres along one vertex, the 600 cells form 15 shells of 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20 and 20 cells. DICTO noticed that this list reads the same backwards. The angles from the vertex to the shells' cell centres mirror it too: 22.24°, 37.76°, 45°, 55.11°, 69.30°, 77.38°, 82.24°, 90°, then 97.76° and on to 157.76°, and each shell's angle and its mirror shell's angle add up to exactly 180°. Both follow from the 600-cell's symmetry through its centre, which carries shell k to shell 14 – k and the vertex to the opposite vertex. Seen in perspective from just beyond the opposite vertex (eye at height –21/10 when the vertex is at 2), the cells of each shell are all of one kind: they have the same six edge lengths, checked exactly. The core is the innermost shell, the 20 cells round the vertex. In this view their common corner is the centre, and their twelve other corners, the vertex's twelve neighbours, form a regular icosahedron about it (§4); so each core cell is the central pyramid of a regular icosahedron, the piece DICTO named the Cleo-Icos (#19). Mosseri and Sadoc flatten the 600-cell into ordinary space to describe icosahedral order in materials (§8); the icosahedron of 20 Cleo-Icos is the 600-cell's vertex cluster brought flat. In DICTO's words, the Cleo-Icos is the 600-cell's original tetrahedron as it must look in flat space: twenty of them fill the regular icosahedron exactly. The shells also explain the vertex-first shadow: its front layer is exactly the core and the next six shells (270 cells), its side-on cells are the middle shell (60), and its back layer is the other seven shells.

DICTO's radial cell projection. In the Polyhedraverse app, DICTO's radial cell projection (RCP) builds the 600-cell cell by cell, each new cell folded across a face of one already placed. It is his rediscovery of the classical reflection construction (Coxeter). The fold between neighbouring cells is the angle with cosine $(1 + 3\sqrt{5})/8$, 15.52°: the same angle as the moment on the vertex–cell turning above. Counted in steps from one cell to a face-to-face neighbour, the 600 cells fall into 16 shells of 1, 4, 12, 24, 36, 52, 68, 76, 78, 72, 64, 56, 40, 12, 4 and 1 cells, checked here on the exact model (schlegel_shells.py). The 600-cell thus unfolds in three ways from one exact model: in DICTO's 16 steps from a cell, in 31 layers of height from a cell, and in the 15 Cleo shells round a vertex.

7. A side note: six impossible things, and Humpty Dumpty's fall

> *“Why, sometimes I've believed as many as six impossible things before breakfast.” — the White Queen, Through the Looking-Glass, ch. 5*

Six of the all-triangle slices are large, with 28 to 72 vertices, and match none of the eight convex deltahedra: the slices with 72 (vertex-first), 32 (edge-first), 30 (face-first), and 48, 42 and 28 (cell-first) vertices. #62 tested each for whether its triangles can all be made equal while its full symmetry is kept. For five of them this is impossible. For the 48-vertex slice it is possible in exactly one way, and the result has a name.

> *“All the King's horses and all the King's men / Couldn't put Humpty Dumpty in his place again.” — Through the Looking-Glass, ch. 6*

The real 48-vertex slice, at height $1 + \sqrt{5}/2$ in the cell-first direction, is **Humpty Dumpty** (DICTO's name): a convex, egg-like solid of 92 triangles, with edge lengths in ratios up to $(5+\sqrt{5})/2 \approx 3.62$ (Figure 5.2). Forced to have all 138 edges equal, it becomes **Fallen Humpty** (placeholder name; Slice-48 in #62): a non-convex solid that folds in and out, with dihedral angles from 81.75° to 243.20°, where the egg was smooth. As DICTO remarked, the rhyme describes the situation exactly: the equal triangles put him together again, but not in his place. What the 600-cell shows is the egg; Fallen Humpty is what our demand for equal triangles makes of it.

> *“When I use a word,” Humpty Dumpty said in rather a scornful tone, “it means just what I choose it to mean—neither more nor less.” — ch. 6*

8. Prior art and credit

Alicia Boole Stott (1900; encouraged by P. H. Schoute) studied series of three-dimensional sections of the regular four-dimensional polytopes and made complete sets of models of the sections of the 600-cell and the 120-cell; two sets of 600-cell section models are kept at the University of Groningen, and a further set is at the Faulkes Institute, Cambridge. Coxeter's *Regular Polytopes* treats the polytopes and the vertex-first sections. We have now read her paper, "On certain series of sections of the regular four-dimensional hypersolids" (Verhandelingen der Koninklijke Akademie van Wetenschappen te Amsterdam, Eerste Sectie, 7, no. 3, 1900, with 5 plates). Her series for the 600-cell is cell-first: she cuts by spaces parallel to one bounding tetrahedron, starting with that tetrahedron and passing through the layers of vertices to the middle, after which "the other sections are now repeated in reverse order"; she finds each section by construction, draws it as a flat plan with its faces coloured by kind of tetrahedron (her diagrams I–VII), builds it as a model, and counts the 600 tetrahedra by colour across the series (her §17). This matches our eight cell-first cuts through layers of corners, from the tip to the centre. Her series has no cuts inside the gaps between layers and no vertex-, edge- or face-first series of the 600-cell, and she does not treat shadows. Her work is the precedent for this atlas, and its cell-first sections at the corner layers are hers. Mosseri and Sadoc flatten the 600-cell into ordinary space to describe icosahedral order in materials (C. R. Physique 15, 2014, doi:10.1016/j.crhy.2013.09.006); that its cells then become Cleo-Icos was noted in the Cleo-Icos reading study of 14 October 2026, made on DICTO's idea. Counts of cells by layer round a vertex of the 600-cell are a natural tabulation and may be known; we did not find them in the sources checked. We did not find in the sources checked: an exact atlas in all four directions including the gaps between vertex heights (Boole Stott's series is the cell-first one through the corner layers), the list of all-triangle slices, or Humpty Dumpty.

9. Conclusion

The 600-cell has 62 slice patterns between its centre and its tips in its four main directions (the far half mirrors the near half), all computed exactly here; exactly twelve are made only of triangles, and each is identified. With the four shadows this is a complete, proved atlas of the White Queen's solid, every slice shown as it really is. Its shadows resolve into exact halves, and its 600 cells stand in 15 shells round one vertex, each shell of one kind of piece, the innermost a core of 20 Cleo-Icos. Its most graceful slice is Humpty Dumpty, the 48-vertex egg. Forcing

equal edges, the aim of #62, succeeds for that one large slice only, and its result, Fallen Humpty, is less graceful than the egg it came from.

Appendix. The 62 slice patterns

Every pattern on the near side, from the centre (height 0) to the tip, in each direction; heights are x·d with d as in §2, exact in $\mathbb{Q}(\sqrt{5})$ ($\sqrt{5}$ written as a factor). A cut through a layer of corners is marked *corners*, a cut inside the gap between two layers *gap* (taken at the gap's midpoint; by the lemma every height in the gap gives the same pattern). V, E, F count the corners, edges and faces; faces are given as number × number of sides. The same table, with edge lengths and dihedral angles, is *data/atlas600.json* in the study attached to this record.

#	direction	cut	height	V	E	F	faces	all triangles
1	vertex-first	corners	0	42	120	80	80×3	yes
2	vertex-first	gap	-1/2+1/2√5	132	330	200	140×3, 60×4	
3	vertex-first	corners	-1+√5	72	210	140	140×3	yes
4	vertex-first	gap	1/2+1/2√5	132	330	200	140×3, 60×4	
5	vertex-first	corners	2	32	90	60	60×3	yes
6	vertex-first	gap	3/2+1/2√5	72	180	110	80×3, 30×4	
7	vertex-first	corners	1+√5	12	30	20	20×3	yes
8	vertex-first	gap	5/2+1/2√5	12	30	20	20×3	yes
9	vertex-first	corners	4	tip: a point (the vertex)				
10	edge-first	corners	0	92	250	160	140×3, 20×4	
11	edge-first	gap	-1/4+1/4√5	142	355	215	150×3, 65×4	
12	edge-first	corners	-1/2+1/2√5	92	255	165	150×3, 15×4	
13	edge-first	gap	1/4+1/4√5	142	355	215	150×3, 65×4	
14	edge-first	corners	1	92	245	155	130×3, 25×4	
15	edge-first	gap	3/4+1/4√5	122	305	185	130×3, 55×4	
16	edge-first	corners	1/2+1/2√5	72	200	130	120×3, 10×4	
17	edge-first	gap	5/4+1/4√5	112	280	170	120×3, 50×4	
18	edge-first	corners	2	77	205	130	110×3, 20×4	
19	edge-first	gap	1+1/2√5	92	230	140	100×3, 40×4	
20	edge-first	corners	√5	82	210	130	100×3, 30×4	
21	edge-first	gap	3/4+3/4√5	92	230	140	100×3, 40×4	
22	edge-first	corners	3/2+1/2√5	32	90	60	60×3	yes
23	edge-first	gap	5/4+3/4√5	52	130	80	60×3, 20×4	
24	edge-first	corners	1+√5	17	45	30	30×3	yes
25	edge-first	gap	7/4+3/4√5	22	55	35	30×3, 5×4	
26	edge-first	corners	5/2+1/2√5	tip: a segment (the edge)				
27	face-first	corners	0	104	282	180	156×3, 24×4	
28	face-first	gap	-1/6+1/6√5	144	360	218	152×3, 66×4	
29	face-first	corners	-1/3+1/3√5	114	300	188	152×3, 36×4	
30	face-first	gap	1/6+1/6√5	144	360	218	152×3, 66×4	
31	face-first	corners	2/3	114	300	188	152×3, 36×4	
32	face-first	gap	1/2+1/6√5	138	345	209	146×3, 63×4	
33	face-first	corners	1/3+1/3√5	78	219	143	134×3, 9×4	
34	face-first	gap	5/6+1/6√5	126	315	191	134×3, 57×4	
35	face-first	corners	4/3	105	270	167	128×3, 39×4	
36	face-first	gap	7/6+1/6√5	114	285	173	122×3, 51×4	
37	face-first	corners	1+1/3√5	84	219	137	110×3, 27×4	
38	face-first	gap	5/6+1/2√5	102	255	155	110×3, 45×4	
39	face-first	corners	2/3+2/3√5	72	192	122	104×3, 18×4	
40	face-first	gap	7/6+1/2√5	96	240	146	104×3, 42×4	
41	face-first	corners	5/3+1/3√5	54	150	98	92×3, 6×4	
42	face-first	gap	3/2+1/2√5	72	180	110	80×3, 30×4	
43	face-first	corners	4/3+2/3√5	30	84	56	56×3	yes
44	face-first	gap	7/6+5/6√5	42	105	65	50×3, 15×4	

45	face-first	corners	1+√5	26	69	45	42×3, 3×4	
46	face-first	gap	3/2+5/6√5	30	75	47	38×3, 9×4	
47	face-first	corners	2+2/3√5	tip: a triangle				
48	cell-first	corners	0	84	240	158	152×3, 6×4	
49	cell-first	gap	1/4	144	360	218	152×3, 66×4	
50	cell-first	corners	1/2	124	324	202	160×3, 42×4	
51	cell-first	gap	3/8+1/8√5	144	360	218	152×3, 66×4	
52	cell-first	corners	1/4+1/4√5	84	234	152	140×3, 12×4	
53	cell-first	gap	1/2+1/4√5	132	330	200	140×3, 60×4	
54	cell-first	corners	3/4+1/4√5	72	198	128	116×3, 12×4	
55	cell-first	gap	7/8+3/8√5	108	270	164	116×3, 48×4	
56	cell-first	corners	1+1/2√5	48	138	92	92×3	yes
57	cell-first	gap	5/4+1/2√5	84	210	128	92×3, 36×4	
58	cell-first	corners	3/2+1/2√5	42	120	80	80×3	yes
59	cell-first	gap	11/8+5/8√5	60	150	92	68×3, 24×4	
60	cell-first	corners	5/4+3/4√5	28	78	52	52×3	yes
61	cell-first	gap	3/2+3/4√5	36	90	56	44×3, 12×4	
62	cell-first	corners	7/4+3/4√5	4	6	4	4×3	yes

Chapter 6. The Red Queen 120-cell

From paper #65, DICTO DOI 10.5281/zenodo.23716140

Aristotle's five pyramids as the cores of the regular four-dimensional solids

Abstract. Aristotle wrote that among solids the pyramid fills space, and the pyramids cut from the centres of the five Platonic solids have been argued over ever since. We prove that **all five of these pyramids are the cores of the regular four-dimensional solids**, seen in perspective: the tetrahedron, octahedron and icosahedron pyramids are the shadows of the cells round one corner of the 5-cell, the 16-cell and the 600-cell, and the cube, octahedron and dodecahedron pyramids are the shadows of the sides of the four-dimensional pyramid over one cell of the tesseract, the 24-cell and the 120-cell. Each shadow has exactly the proportions of Aristotle's pyramid, and the five come in polar pairs. The 120-cell, polar of the White Queen 600-cell of #63 and named the Red Queen by DICTO, is built exactly from the White Queen: its 600 corners are her 600 cell centres, its corners stand in her fifteen Cleo shells, its 120 dodecahedra in nine layers round one, and its core is the dodecahedron cut into twelve pentagonal pyramids where hers is the icosahedron cut into twenty Cleo-Icos. No convex regular solid has dodecahedra round a corner, but the star polytope {5/2,5,3} does: its twelve golden spikes at a corner cast exactly Aristotle's dodecahedron pyramids. Her full atlas is exact: 124 slice patterns in the four main directions, passing cell-first through the Archimedean solids of the icosahedral family, and four shadows, each filled exactly once by its front cells and once by its back cells, each the polar of the White Queen's middle slice.

1. From the White Queen to the Red Queen

In #63, the White Queen 600-cell, 600 regular tetrahedra stand round 120 corners, and round one corner they fall into fifteen shells whose core is twenty Cleo-Icos filling a regular icosahedron (#19, #44). DICTO proposed her polar swap: every corner of one solid becomes a cell of the other. The polar of the 600-cell is the 120-cell, 120 regular dodecahedra round 600 corners. DICTO named it the Red Queen, after the other queen of *Through the Looking-Glass*. He then proposed that the dodecahedral pyramid, the one of Aristotle's five pyramids the White Queen lacks, sits at the Red Queen's core, and that all five pyramids are four-dimensional shadows. They are.

2. The Red Queen, built from the White Queen

We build the 120-cell from the exact model of #63, in $\mathbb{Q}(\sqrt{5})$. Its corners are the centres of the White Queen's 600 cells; two corners are joined when their cells share a triangle (1200 edges); the five cells round each White Queen edge give a pentagon (720 pentagons); the twenty cells round each White Queen corner give a cell (120 cells). Checked exactly: all corners on one sphere, all edges equal, four edges at every corner, every pentagon regular and shared by exactly two cells, every cell a regular dodecahedron (13 of 13 checks).

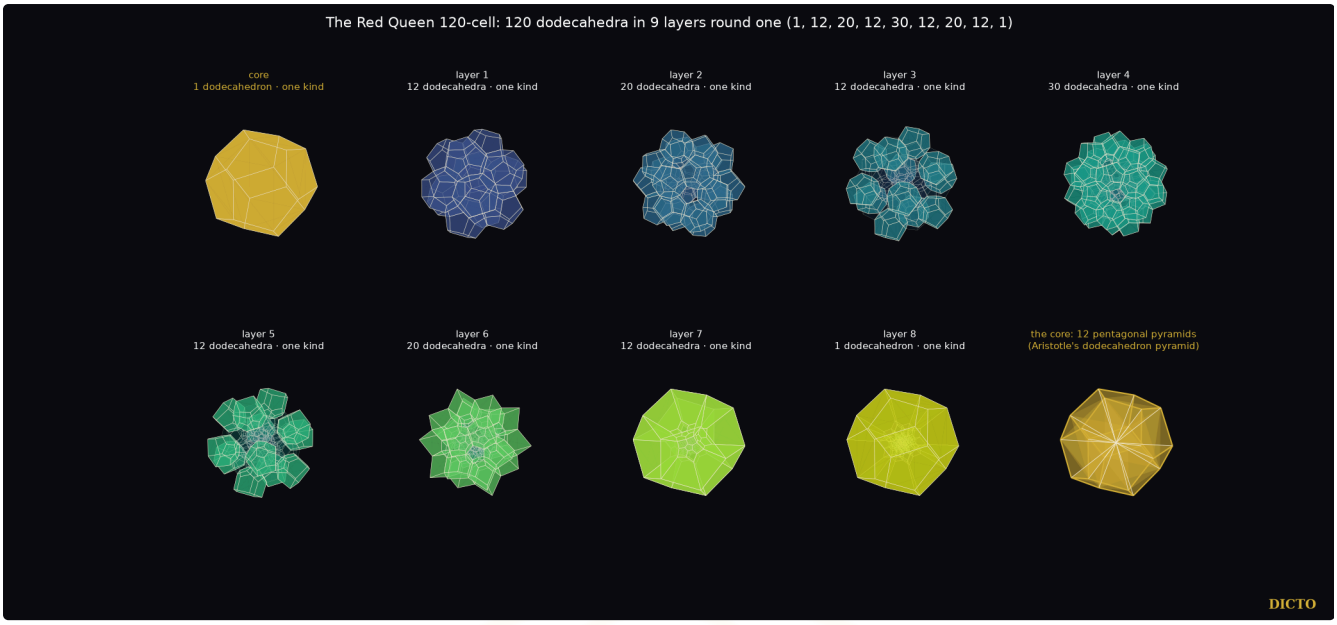


Figure 6.1. The Red Queen: 120 dodecahedra in nine layers round one (1, 12, 20, 12, 30, 12, 20, 12, 1), seen in perspective from just beyond the opposite cell, each layer one kind of piece; last panel: the gold core cut from its centre into its twelve pentagonal pyramids.

Layers and core. Round one dodecahedron the 120 cells stand in nine layers of 1, 12, 20, 12, 30, 12, 20, 12 and 1, each one kind of piece in the perspective view, and the 600 corners in fifteen layers of 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20 and 20: exactly the Cleo shells of the White Queen, since her cell centres are the Red Queen's corners. Both lists are palindromes, the same forwards and backwards, as the White Queen's shells are (#63): the Red Queen is centrally symmetric, so the layers counted from one cell match, in reverse, the layers counted from the opposite cell. The two cores exchange one for one: each of the White Queen's twenty Cleo-Icos becomes one of the twenty corners of the Red Queen's central dodecahedron, and each of the twelve corners of the White Queen's icosahedron becomes one of its twelve pentagons. The White Queen's twenty Cleo-Icos are real cells, the shadows of the twenty tetrahedra at her corner. The Red Queen's core is one whole cell; its twelve pentagonal pyramids, Aristotle's dodecahedron pyramid, appear when she is divided from her centre into 120 four-dimensional pyramids, one over each cell: the twelve sides of the central one cast them exactly (§3). In the star Red Queen (§5) they are parts of real cells again.

How it closes. Three regular dodecahedra round an edge fill 3 × 116.57° = 349.7° and leave a gap of 10.3°, so dodecahedra cannot close round an edge in ordinary space; five regular tetrahedra leave 7.36°. As three pentagons round a corner close only by folding up into the dodecahedron, these gaps close by bending into the fourth dimension, and the Red Queen closes with three dodecahedra round every edge. In the perspective view the pieces close with no gaps and no overlaps, slightly bent, and the last dodecahedron becomes the outside, as the last pentagon of a flattened dodecahedron becomes its outer rim.

3. Main theorem: Aristotle's five pyramids

> **Theorem.** In perspective from just beyond the far side, along a corner or along a cell centre:
>
> (1) **Corner cores.** The cells at one corner of the 5-cell, the 16-cell and the 600-cell cast the face pyramids of the regular tetrahedron (4), octahedron (8) and icosahedron (20, the Cleo-Icos), apex at the centre.
>
> (2) **Cell cores.** The sides of the four-dimensional pyramid from the solid's centre over one cell cast the face pyramids of that cell: tetrahedron (5-cell, 16-cell, 600-cell), cube (tesseract, 6), octahedron (24-cell, 8) and dodecahedron (120-cell, 12).

(3) **Exact proportions.** In every case the shadow is exactly Aristotle's face pyramid of that Platonic solid: its side edge² over its base edge² is 3/8, 1/2, 3/4, (5 + √5)/8 and (9 + 3√5)/8 for the tetrahedron, octahedron, cube, icosahedron and dodecahedron.

(4) **Polar pairs.** 16-cell (8 octahedron pyramids) ↔ tesseract (6 cube pyramids); 600-cell (20 Cleo-Icos) ↔ 120-cell (12 pentagonal pyramids); the 5-cell and the 24-cell are their own polars.

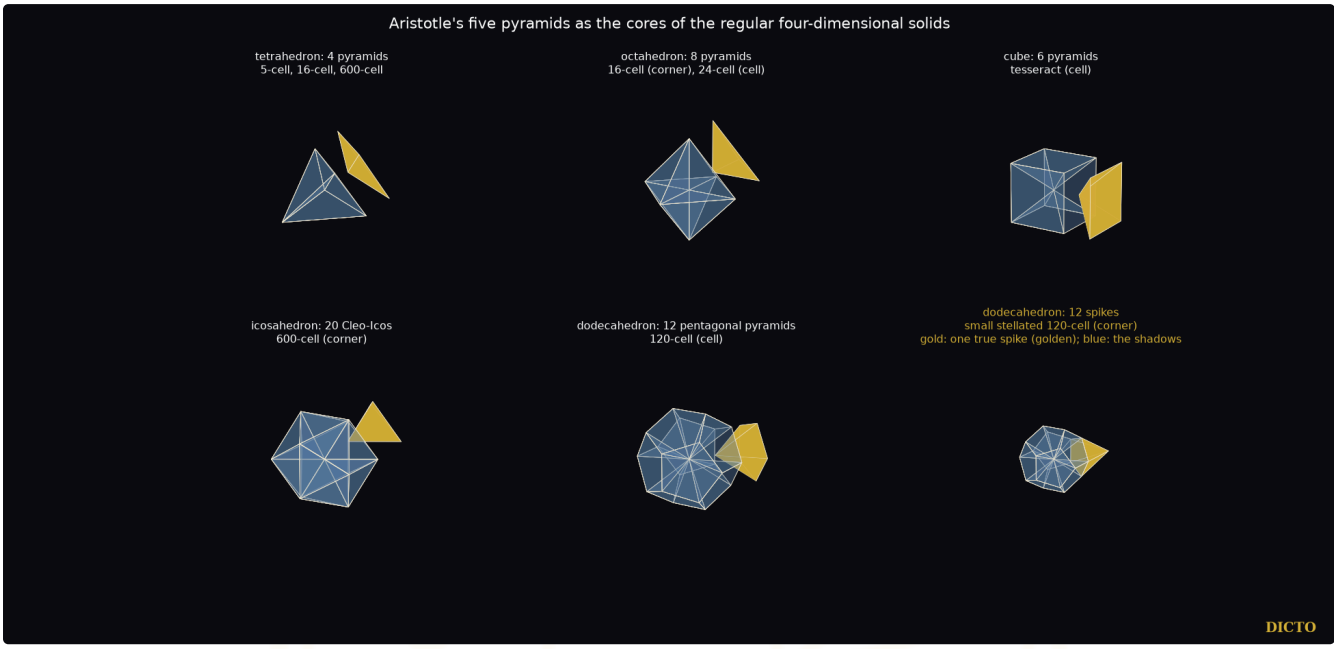


Figure 6.2. Aristotle's five pyramids as the cores of the regular four-dimensional solids, one pyramid of each drawn out in gold. Last panel: the star 120-cell (§5), one true spike in gold, its shadows in blue.

Proof. The six solids are built exactly (the 5-cell in the hyperplane of five coordinates summing to zero; the 16-cell, tesseract and 24-cell from their standard coordinates; the 600-cell and 120-cell as in §2). For each we take the eye on the axis just beyond the far side and project. On the axis the corner, or the cell centre and the solid's centre, fall on the centre of the picture, so every piece through it becomes a pyramid with its apex there; we check exactly that the far ends of these pieces form a regular solid centred on the picture's centre, count its faces, and compare side edge² over base edge² with Aristotle's pyramid. All ten cases agree exactly (10 of 10).

4. True pieces and their shadows

DICTO pointed out that the pyramids in four dimensions may have different proportions from their shadows. They do. The corner cores are regular tetrahedra in four dimensions, side edge equal to base edge, but their shadows are the flatter face pyramids. The sides of a cell core run from the solid's centre to its corners, so in four dimensions their side edge² over base edge² is the solid's radius² over edge²: 2/5 (5-cell), 1/2 (16-cell), 1 (tesseract and 24-cell), (3 + √5)/2 (600-cell) and 7 + 3√5 (120-cell), while their shadows are again Aristotle's. The four-dimensional solids carry tall pieces; their shadows, in our space, are Aristotle's five pyramids.

5. The star core

No convex regular four-dimensional solid has dodecahedra round a corner: that would be {3,5,3}, which never closes and is the infinite icosahedral honeycomb of hyperbolic space. The star polytope {5/2,5,3}, the small stellated 120-cell of Hess, does. It stands on the White Queen's 120 corners, with 1200 edges joining corners one radius apart, 720 pentagrams and 120 small stellated dodecahedra, and round every corner a regular dodecahedron of twelve cells. Each of these twelve cells meets the corner with one spike, a pentagonal pyramid whose side edge is exactly φ times its base edge. In perspective the twelve spikes cast exactly the dodecahedron cut into Aristotle's twelve pentagonal pyramids (checked exactly, side edge² over base edge² = (9

+ 3√5)/8). So the dodecahedral pyramids that no convex solid can hold at a corner sit at the corner of the star Red Queen: golden in four dimensions, Aristotle's in their shadow.

6. The Red Queen's atlas

As for the White Queen in #63, we cut the Red Queen in her four main directions, through a corner, an edge, a pentagon and a cell, from the centre to the tip, at every corner height and once inside every gap, exactly in $\mathbb{Q}(\sqrt{5})$. The one-pattern-per-gap lemma of #63 holds here too (cuts at ¼, ½ and ¾ of every gap cross the same edges), so this list is complete. Her corners are the centres of the White Queen's cells in the coordinates of #63 (corner radius² 7/4 + 3√5/4, edge 1/2). The four directions d are the average of the corners of one element: corner-first d = ((7+3√5)/8, (3+√5)/8, 0, 1/4), a corner; edge-first d = ((7+3√5)/8, (3+√5)/8, 0, 0), the midpoint of an edge; pentagon-first d = ((7+3√5)/8, (5+2√5)/20, (5+3√5)/40, 0), the centre of a pentagon; cell-first d = ((7+3√5)/8, 0, 0, 0), the centre of a cell. They are exactly parallel to the White Queen's cell-, triangle-, edge- and vertex-first directions. Every slice is checked exactly as in #63: repeated points merged by exact comparison, every face plane tested to support the slice by an exact sign in $\mathbb{Q}(\sqrt{5})$, every face ordered by an exact two-dimensional hull and turned outward, every edge in exactly two faces, and V − E + F = 2. The atlas adds 17 checks of its own, all passed: among them that each gap gives the same faces when cut at ¼, ½ and ¾ of its height, that inside a gap each slice face is the cross-section of exactly one dodecahedron, that every cut point is a corner of its slice, that each shadow is tiled exactly once by its front cells and once by its back cells (exact volume), and that the polar pairings with the White Queen hold.

> **Atlas.** The Red Queen has **124 slice patterns**: 31 corner-first (faces of 3 to 8 sides, largest 108 vertices), 45 edge-first (3 to 9, the only nine-sided faces, largest 114), 33 pentagon-first (3 to 8 and 10, largest 120 vertices and 62 faces) and 15 cell-first (faces of 3, 4, 5, 6, 8 and 10 sides, largest 120 vertices).
>
> Cell-first she passes through the Archimedean solids of the icosahedral family, each exact: the rhombicosidodecahedron, the truncated icosahedron and the truncated dodecahedron (each at one height inside its gap), the icosidodecahedron, and regular dodecahedra (edge φ times the cell's) up to the tip, where the slice is the cell itself; these dodecahedra are her only all-pentagon slices. The other directions begin with a regular tetrahedron (corner-first), a regular triangular prism (edge-first) and a regular pentagonal prism (pentagon-first), and regular decagons appear in the pentagon-first middle slice and in one cell-first gap.



Figure 6.3. The Red Queen's atlas: her slice patterns in the four main directions, from the centre to the tip.

Shadows. Her four shadows are

Direction	Vertices · edges · faces	Faces	Front · back · side-on cells
corner-first	158 · 240 · 84	72 pentagons, 12 decagons	54 · 54 · 12
edge-first	180 · 282 · 104	96 pentagons, 6 decagons, 2 twelve-sided	56 · 56 · 8
pentagon-first	160 · 250 · 92	80 pentagons, 10 octagons, 2 regular decagons	54 · 54 · 12
cell-first	80 · 120 · 42	12 pentagons, 30 hexagons	45 · 45 · 30

Every face that is not a pentagon is a side-on cell seen flat. As in the White Queen, each shadow resolves into two exact halves: the front cells fill it exactly once and so do the back cells (exact volume). And the two queens' atlases mirror each other by polarity: each of the Red Queen's shadows is the polar of the White Queen's middle slice in the same direction, and each of her middle slices is the polar of the White Queen's shadow.

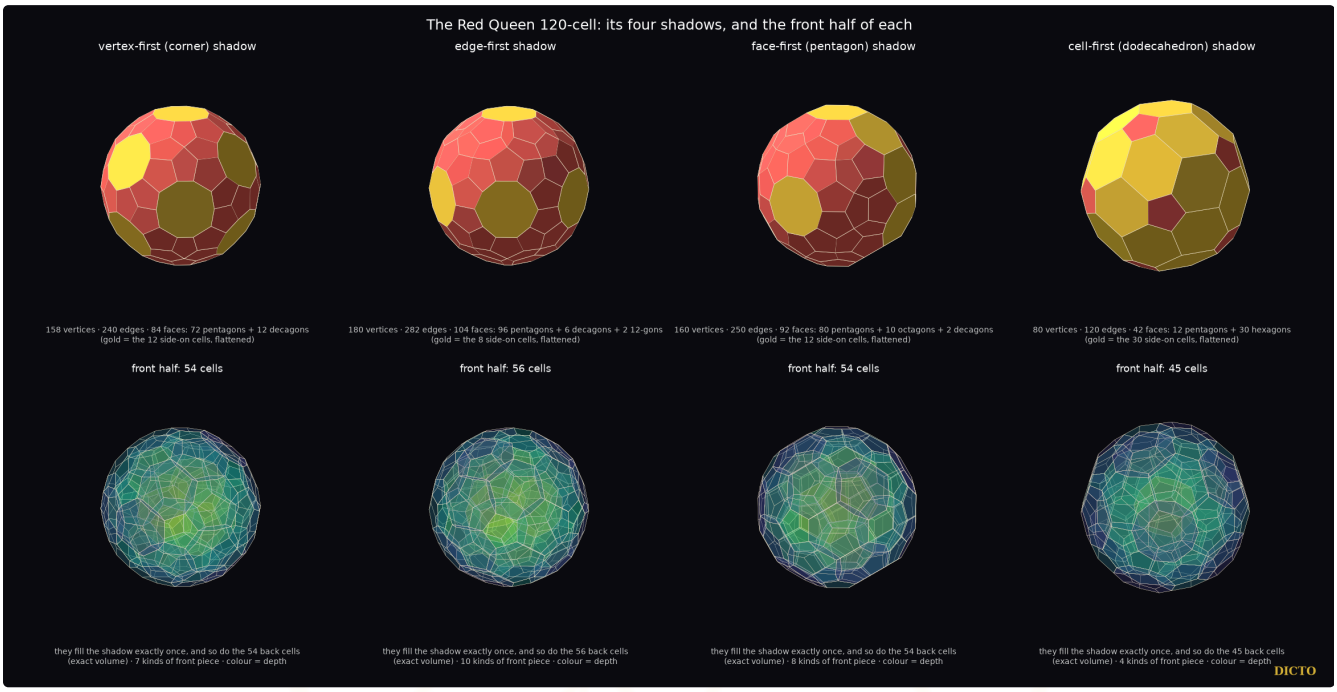


Figure 6.4. The Red Queen's four shadows (gold: side-on cells, seen flat) and the front half of each, which fills its shadow exactly once.

7. Prior art and credit

The 120-cell and 600-cell are Schläfli's (about 1852, published 1901); their coordinates, polarity and the star polytopes {5/2,5,3} and the others are in Coxeter's *Regular Polytopes*, the star polytopes first found by Schläfli and Hess (1883). Alicia Boole Stott, in "On certain series of sections of the regular four-dimensional hypersolids" (Verhandelingen der Koninklijke Akademie van Wetenschappen te Amsterdam, Eerste Sectie, 7, no. 3, 1900), cut the 120-cell cell-first, parallel to one bounding dodecahedron, through the layers of corners to the middle, and drew each section as a flat plan with coloured faces (her diagrams VIII–XIV); we read her paper for this version. Her series matches our eight cell-first cuts through corner layers, from the cell to the centre, among them the rhombicosidodecahedron and the icosidodecahedron, which are hers. The truncated icosahedron and the truncated dodecahedron lie inside gaps between corner layers, which her series does not cut, and she has no corner-, edge- or pentagon-first series and no shadows. The layers 1, 12, 20, 12, 30, … of the 120-cell's cells round one cell are known, for example in Schleimer and Segerman's *Puzzling the 120-cell*. Aristotle's claim and the long dispute over how many pyramids meet round a point are traced by Lagarias and Zong (2012); the face pyramids of the Platonic solids and which of them fill space are #44, *Aristotle's Pyramid*, and the Cleo-Icos is #19. We found no earlier statement that Aristotle's five pyramids are exactly the perspective cores of the regular four-dimensional solids, nor of the star core of {5/2,5,3}.

8. Conclusion

Aristotle's five pyramids are the shadows of the four-dimensional regular solids' cores: three as the cells round a corner, three as the sides of the pyramid over a cell, each with exactly Aristotle's proportions, paired by polarity. The Red Queen, built from the White Queen, holds the dodecahedral one at her core, and the star Red Queen holds it at every corner, golden in four dimensions. Her atlas of 124 slices and four shadows mirrors the White Queen's by polarity, shadow for middle slice.

Appendix. The 124 slice patterns

Every pattern from the centre (height 0) to the tip, in each direction; heights are x·d with d as in §6, exact in $\mathbb{Q}(\sqrt{5})$ (√5 written as a factor). *corners* marks a cut through a layer of corners, *gap* a cut inside the gap between two layers (by the one-pattern-per-gap lemma every height in the gap gives the same pattern). V, E, F count corners, edges and faces; faces are given as number × number of sides; the last column names the uniform solids. The same table, with edge lengths and dihedral angles, is *data/atlas120.json* in the study attached to this record.

#	direction	cut	height	V	E	F	faces	uniform solid
1	corner-first	corners	0	78	120	44	24×5, 20×6	
2	corner-first	gap	3/32+1/32√5	108	162	56	12×4, 12×5, 20×6, 12×8	
3	corner-first	corners	3/16+1/16√5	84	138	56	12×3, 12×4, 32×6	
4	corner-first	gap	7/32+3/32√5	108	162	56	12×3, 32×6, 12×8	
5	corner-first	corners	1/4+1/8√5	72	114	44	4×3, 24×5, 16×6	
6	corner-first	gap	11/32+5/32√5	84	126	44	4×3, 12×5, 16×6, 12×7	
7	corner-first	corners	7/16+3/16√5	76	114	40	12×5, 28×6	
8	corner-first	gap	15/32+7/32√5	100	150	52	12×4, 12×5, 16×6, 12×8	
9	corner-first	corners	1/2+1/4√5	76	126	52	12×3, 12×4, 28×6	
10	corner-first	gap	9/16+1/4√5	100	150	52	12×3, 28×6, 12×8	
11	corner-first	corners	5/8+1/4√5	76	114	40	12×5, 28×6	
12	corner-first	gap	21/32+9/32√5	76	114	40	12×5, 28×6	
13	corner-first	corners	11/16+5/16√5	64	102	40	4×3, 24×5, 12×6	
14	corner-first	gap	25/32+11/32√5	88	132	46	4×3, 6×4, 12×5, 24×7	
15	corner-first	corners	7/8+3/8√5	56	96	42	12×3, 6×4, 12×5, 12×6	
16	corner-first	gap	29/32+13/32√5	88	132	46	16×3, 12×6, 18×8	
17	corner-first	corners	15/16+7/16√5	52	84	34	4×3, 12×4, 18×6	
18	corner-first	gap	33/32+15/32√5	64	96	34	12×4, 10×6, 12×7	
19	corner-first	corners	9/8+1/2√5	40	60	22	12×5, 10×6	
20	corner-first	gap	37/32+17/32√5	40	60	22	12×5, 10×6	
21	corner-first	corners	19/16+9/16√5	40	60	22	12×5, 10×6	
22	corner-first	gap	5/4+9/16√5	48	72	26	4×3, 22×6	
23	corner-first	corners	21/16+9/16√5	36	60	26	4×3, 12×4, 10×6	
24	corner-first	gap	43/32+19/32√5	48	72	26	12×4, 8×6, 6×8	
25	corner-first	corners	11/8+5/8√5	24	36	14	6×4, 8×6	
26	corner-first	gap	47/32+21/32√5	24	36	14	6×4, 8×6	
27	corner-first	corners	25/16+11/16√5	12	18	8	4×3, 4×6	
28	corner-first	gap	51/32+23/32√5	12	18	8	4×3, 4×6	
29	corner-first	corners	13/8+3/4√5	4	6	4	4×3	regular tetrahedron
30	corner-first	gap	27/16+3/4√5	4	6	4	4×3	regular tetrahedron
31	corner-first	corners	7/4+3/4√5	tip: a corner				
32	edge-first	corners	0	96	150	56	24×4, 26×6, 6×8	
33	edge-first	gap	3/64+1/64√5	114	171	59	15×4, 12×5, 20×6, 6×8, 6×9	
34	edge-first	corners	3/32+1/32√5	102	159	59	12×3, 3×4, 12×5, 20×6, 6×7, 6×8	
35	edge-first	gap	7/64+3/64√5	114	171	59	12×3, 3×4, 26×6, 6×7, 12×8	
36	edge-first	corners	1/8+1/16√5	84	129	47	3×4, 24×5, 14×6, 6×7	
37	edge-first	gap	11/64+5/64√5	90	135	47	3×4, 12×5, 26×6, 6×7	
38	edge-first	corners	7/32+3/32√5	78	123	47	6×3, 3×4, 12×5, 26×6	
39	edge-first	gap	9/32+1/8√5	102	153	53	6×3, 6×4, 12×5, 8×6, 12×7, 9×8	
40	edge-first	corners	11/32+5/32√5	78	123	47	6×3, 6×4, 6×5, 29×6	
41	edge-first	gap	25/64+11/64√5	90	135	47	6×3, 6×5, 23×6, 12×7	
42	edge-first	corners	7/16+3/16√5	78	117	41	12×5, 29×6	
43	edge-first	gap	29/64+13/64√5	90	135	47	6×3, 6×5, 29×6, 6×8	
44	edge-first	corners	15/32+7/32√5	72	117	47	8×3, 6×4, 12×5, 21×6	

45	edge-first	gap	33/64+15/64√5	90	135	47	2×3, 6×4, 12×5, 9×6, 18×7	
46	edge-first	corners	9/16+1/4√5	78	117	41	2×3, 12×5, 21×6, 6×7	
47	edge-first	gap	39/64+17/64√5	90	135	47	2×3, 6×4, 12×5, 9×6, 18×7	
48	edge-first	corners	21/32+9/32√5	86	129	45	6×4, 12×5, 15×6, 12×7	
49	edge-first	gap	43/64+19/64√5	86	129	45	6×4, 12×5, 15×6, 12×7	
50	edge-first	corners	11/16+5/16√5	71	114	45	6×3, 6×4, 12×5, 21×6	
51	edge-first	gap	47/64+21/64√5	86	129	45	6×3, 6×5, 27×6, 6×8	
52	edge-first	corners	25/32+11/32√5	62	99	39	6×3, 18×5, 15×6	
53	edge-first	gap	51/64+23/64√5	86	129	45	12×3, 18×6, 6×7, 9×8	
54	edge-first	corners	13/16+3/8√5	80	123	45	12×3, 12×5, 6×6, 6×7, 9×8	
55	edge-first	gap	55/64+25/64√5	86	129	45	6×3, 6×4, 12×5, 12×7, 9×8	
56	edge-first	corners	29/32+13/32√5	62	99	39	21×4, 12×6, 6×7	
57	edge-first	gap	61/64+27/64√5	74	111	39	15×4, 12×6, 6×7, 6×8	
58	edge-first	corners	1+7/16√5	62	99	39	12×3, 3×4, 6×5, 12×6, 6×8	
59	edge-first	gap	65/64+29/64√5	74	111	39	12×3, 3×4, 6×5, 18×8	
60	edge-first	corners	33/32+15/32√5	50	75	27	3×4, 6×5, 18×6	
61	edge-first	gap	69/64+31/64√5	54	81	29	2×3, 3×4, 24×6	
62	edge-first	corners	9/8+1/2√5	42	66	26	2×3, 6×4, 6×5, 12×6	
63	edge-first	gap	19/16+17/32√5	48	72	26	6×4, 6×5, 8×6, 6×7	
64	edge-first	corners	5/4+9/16√5	36	54	20	12×5, 8×6	
65	edge-first	gap	83/64+37/64√5	42	63	23	3×4, 6×5, 14×6	
66	edge-first	corners	43/32+19/32√5	30	51	23	6×3, 9×4, 8×6	
67	edge-first	gap	87/64+39/64√5	42	63	23	6×3, 6×4, 2×6, 9×8	
68	edge-first	corners	11/8+5/8√5	30	45	17	6×4, 11×6	
69	edge-first	gap	91/64+41/64√5	30	45	17	6×4, 11×6	
70	edge-first	corners	47/32+21/32√5	18	27	11	6×4, 5×6	
71	edge-first	gap	97/64+43/64√5	18	27	11	6×4, 5×6	
72	edge-first	corners	25/16+11/16√5	15	24	11	6×3, 5×6	
73	edge-first	gap	101/64+45/64√5	18	27	11	6×3, 2×6, 3×8	
74	edge-first	corners	51/32+23/32√5	6	9	5	2×3, 3×4	regular triangular prism
75	edge-first	gap	105/64+47/64√5	6	9	5	2×3, 3×4	
76	edge-first	corners	27/16+3/4√5	tip: an edge				
77	pentagon-first	corners	0	80	130	52	20×4, 20×5, 10×6, 2×10	
78	pentagon-first	gap	1/16+1/40√5	100	150	52	10×4, 10×5, 20×6, 10×7, 2×10	
79	pentagon-first	corners	1/8+1/20√5	90	140	52	10×3, 12×5, 20×6, 10×7	
80	pentagon-first	gap	5/32+11/160√5	120	180	62	20×3, 2×5, 10×6, 10×7, 20×8	
81	pentagon-first	corners	3/16+7/80√5	80	130	52	10×3, 10×4, 2×5, 30×6	
82	pentagon-first	gap	1/4+9/80√5	100	150	52	10×4, 12×5, 20×6, 10×8	
83	pentagon-first	corners	5/16+11/80√5	80	120	42	12×5, 30×6	
84	pentagon-first	gap	11/32+5/32√5	90	135	47	5×4, 12×5, 20×6, 10×7	
85	pentagon-first	corners	3/8+7/40√5	85	130	47	5×4, 22×5, 10×6, 10×7	
86	pentagon-first	gap	7/16+1/5√5	90	135	47	5×4, 12×5, 20×6, 10×7	
87	pentagon-first	corners	1/2+9/40√5	70	115	47	10×3, 5×4, 12×5, 20×6	
88	pentagon-first	gap	9/16+1/4√5	90	135	47	10×3, 12×5, 10×6, 15×8	
89	pentagon-first	corners	5/8+11/40√5	70	105	37	12×5, 25×6	

90	pentagon-first	gap	21/32+47/160√5	90	135	47	10×3, 2×5, 25×6, 10×8	
91	pentagon-first	corners	11/16+5/16√5	70	115	47	10×3, 10×4, 2×5, 25×6	
92	pentagon-first	gap	3/4+27/80√5	90	135	47	10×4, 12×5, 15×6, 10×8	
93	pentagon-first	corners	13/16+29/80√5	60	95	37	10×4, 12×5, 15×6	
94	pentagon-first	gap	7/8+31/80√5	70	105	37	10×4, 25×6, 2×10	
95	pentagon-first	corners	15/16+33/80√5	60	95	37	10×3, 10×5, 15×6, 2×10	
96	pentagon-first	gap	31/32+69/160√5	70	105	37	10×3, 10×5, 5×6, 10×8, 2×10	
97	pentagon-first	corners	1+9/20√5	50	75	27	12×5, 15×6	
98	pentagon-first	gap	17/16+19/40√5	60	90	32	5×4, 12×5, 10×6, 5×8	
99	pentagon-first	corners	9/8+1/2√5	40	70	32	10×3, 10×4, 2×5, 10×6	
100	pentagon-first	gap	37/32+83/160√5	60	90	32	10×3, 5×4, 2×5, 15×8	
101	pentagon-first	corners	19/16+43/80√5	40	60	22	5×4, 2×5, 15×6	
102	pentagon-first	gap	5/4+9/16√5	40	60	22	5×4, 2×5, 15×6	
103	pentagon-first	corners	21/16+47/80√5	30	45	17	12×5, 5×6	
104	pentagon-first	gap	11/8+49/80√5	30	45	17	12×5, 5×6	
105	pentagon-first	corners	23/16+51/80√5	20	35	17	10×3, 2×5, 5×6	
106	pentagon-first	gap	47/32+21/32√5	30	45	17	10×3, 5×8, 2×10	
107	pentagon-first	corners	3/2+27/40√5	10	15	7	5×4, 2×5	regular pentagonal prism
108	pentagon-first	gap	25/16+7/10√5	10	15	7	5×4, 2×5	
109	pentagon-first	corners	13/8+29/40√5	tip: a pentagon				
110	cell-first	corners	0	80	120	42	12×5, 30×6	
111	cell-first	gap	7/64+3/64√5	80	120	42	12×5, 30×6	
112	cell-first	corners	7/32+3/32√5	80	120	42	12×5, 30×6	
113	cell-first	gap	9/32+1/8√5	120	180	62	20×3, 12×5, 30×8	
114	cell-first	corners	11/32+5/32√5	60	120	62	20×3, 30×4, 12×5	rhombicosidodecahedron
115	cell-first	gap	29/64+13/64√5	120	180	62	30×4, 20×6, 12×10	
116	cell-first	corners	9/16+1/4√5	60	90	32	12×5, 20×6	
117	cell-first	gap	47/64+21/64√5	60	90	32	12×5, 20×6	truncated icosahedron
118	cell-first	corners	29/32+13/32√5	60	90	32	12×5, 20×6	
119	cell-first	gap	65/64+29/64√5	60	90	32	12×5, 20×6	
120	cell-first	corners	9/8+1/2√5	30	60	32	20×3, 12×5	icosidodecahedron
121	cell-first	gap	19/16+17/32√5	60	90	32	20×3, 12×10	truncated dodecahedron
122	cell-first	corners	5/4+9/16√5	20	30	12	12×5	regular dodecahedron
123	cell-first	gap	87/64+39/64√5	20	30	12	12×5	regular dodecahedron
124	cell-first	corners	47/32+21/32√5	20	30	12	12×5	regular dodecahedron

Epilogue. What has been shown

Conclusion

The book set out to see two solids of four-dimensional space, completely and exactly, and to find what ties them to the solids of our own space. The lead-up gave the pieces: three small pieces close the gaps between icosahedra (Chapter 1); of Aristotle's five pyramids only the cube's, and the tetrahedron's with the octahedron's, fill space (Chapter 2); face to face the five keep to their own solids, and glued together DICTO's pieces make no new deltahedron in the range searched (Chapter 3). The shadows then gave the new shapes that gluing could not: variants of the convex deltahedra that join differently, and deltahedra rebuilt with every edge equal (Chapter 4).

The White Queen has been mapped completely: 62 slice patterns in her four main directions, every one exact, twelve made only of triangles; four shadows, each resolving into an exact front half and back half; and her 600 cells in fifteen shells round one corner, the innermost a core of twenty Cleo-Icos, the icosahedron's pyramid (Chapter 5). Alicia Boole Stott's series of 1900 is the first part of this atlas, and its cell-first slices at the corner layers are hers. The Red Queen, built exactly from the White Queen, mirrors her by polarity, shadow for middle slice, and holds the dodecahedron's pyramid at her core. Aristotle's five pyramids are the cores of the four-dimensional regular solids, each with exactly its proportions, in polar pairs (Chapter 6). The pyramid Aristotle said fills space is, seen from four dimensions, the core of a regular solid.

Open problems

(1) Which tilings of space by the tetrahedron's and the octahedron's pyramids exist besides the octet cut up and its rearrangements (Chapter 2)? (2) Do gluings of five or more pieces with the regular solids, or six or more of DICTO's pieces alone, give a new deltahedron, and do icosahedra with Antonius fill space (Chapter 3)? (3) Which of the new deltahedra of Chapter 4 tile space, alone or with others? (4) The atlases of Chapters 5 and 6 cover the four main directions; the slices in other directions are not mapped.

Appendix. Sources by chapter

Each chapter's own list of sources checked, data and code, as published with its paper.

Chapter 1. AXE, CLEO and FUJI (#19, DICTO DOI 10.5281/zenodo.23351291)

Credits

The Lotus seed, the 63-piece build, UNITY, the Chain of Karma, the Stella-Corona and the names AXE, CLEO, FUJI, Hasu, Triskelion, Triskelion Hex, Inner Eye, DESHI and King of Pentacles are DICTO's. AXE and CLEO are the Mosseri–Sadoc tiles *t6* and *t4* [Mosseri & Sadoc 1982; Kramer 1982; Koca et al. 2020]. The DESHI arrangement is R. W. Gray's [Gray 2002]. Study data, scripts and checks: studies/2026-10-10-icosa-13 and studies/2026-10-14-axe-cleo-fuji at https://github.com/DICTOR-Master/kaleidohedra.

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Chapter 2. Aristotle's Pyramid (#44, DICTO DOI 10.5281/zenodo.23529497)

Data and reproducibility

In studies/2026-10-21-aristotle-pyramids: pyramid_checks.py (exact arithmetic in $\mathbb{Q}(\sqrt{5})$ with Python fractions; angles handled through $\cos^2$ and the sign of cos; log pyramid-checks.log, 47 PASS, no FAIL): the solids' dihedral angles; every base and side angle of the five pyramids; the facts used in Lemma 2; the test over all 31 sets; the octet cell and its 8 + 8 pyramids; the slabs of the octet patch; the golden rhombohedron, Cleo-Icos, Antonius and their angles. angle_relations.py (angle-relations.log): the angles to 80 digits and a PSLQ search finding no integer relation among π, α, θ_D, θ_I. Figures: aristotle_figs.py (float copies of the exact solids into the viewer of studies/2026-10-19-five-fold-octet), jewel-shots.mjs (Playwright, 2× pixel ratio) and aristotle_compose.py. The 2,187-piece patch of Cleo-Icos and Antonius is that of [3].

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10. J. C. Lagarias, D. Moews, Polytopes that fill $\mathbb{R}^n$ and scissors congruence, Discrete Comput. Geom. 13 (1995) 573–583, doi:10.1007/BF02574064; acknowledgment of priority, 14 (1995) 359–360, doi:10.1007/BF02570711 (link only; not read).

Appendix. The reference library searched

The library's manifests list 269 entries, of which 94 have a local file; these give the 81 distinct sources with a text layer listed below and 13 page images. The full text of the 81 sources was searched on 21 October 2026 for: Aristotle, De Caelo, On the Heavens, pyramid, fill space, space-filling, tile space, Dehn, dihedral angle, central pyramid, face pyramid, Maurolico, octet, tetrahedral-octahedral, golden rhombohedron, Cleo and Antonius. The relevant hits are Lagarias and Zong [1] (the history above), Conway, Radin and Sadun [4], the Dehn invariants of icosahedral tiles in Ogievetsky and Papadopolos, the tetrahedral-octahedral honeycomb (Wikipedia), and golden rhombohedra in Senechal, Dietl and Eschenburg and Kabai's demonstrations (all in the list below); none treats the face pyramids of the Platonic solids as tiles. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the 175 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
J. H. Conway, C. Radin, L. Sadun	1999	On angles whose squared trigonometric functions are rational
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
K. Bezdek, Z. Lángi, M. Naszódi, P. Papez	2007	Ball-polyhedra
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids

B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
L. Montejano, E. Roldán-Pensado	2017	Meissner polyhedra
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Hynd	2023	The perimeter and volume of a Reuleaux polyhedron

R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)

Chapter 3. The Five Pyramids: Shared Networks and Deltahedra (#0, DICTO DOI)

[1] N. W. Johnson, "Convex polyhedra with regular faces", Canadian Journal of Mathematics 18 (1966) 169–200.
[2] H. Freudenthal and B. L. van der Waerden, "Over een bewering van Euclides", Simon Stevin 25 (1947) 115–121.
[3] Y. Tsuruta, J. Mitani, Y. Kanamori and Y. Fukui, "Enumeration of deltahedral graphs with up to 10 vertices", Proceedings of ICGG 2014.
[4] J. C. Lagarias and C. Zong, "Mysteries in packing regular tetrahedra", Notices of the American Mathematical Society 59 (2012) 1540–1549.
[5] R. Mosseri and J.-F. Sadoc, "Two and three dimensional non-periodic networks obtained from self-similar tiling", in The Physics of Quasicrystals, World Scientific (1987).

Data and reproducibility

DICTO, studies/2026-10-27-five-pyramids: pieces.py, q1_pyramids.py, exact_faces.py, enum_glue.py (run on dicto-big-brother), validate.py, classify.py, render.py; data/q1-pyramids.json, data/face-matching-exact.json, data/q2-deltahedra.json, data/q2-deltahedra-summary.csv; prior-art.md.

References

Chapter 4. Shadows of the Fourth Dimension (#62, DICTO DOI 10.5281/zenodo.23687223)

Sources checked (27 October 2026; full record: study folder 2026-10-27-4d-shadows, prior-art.md, and 2026-10-27-five-pyramids, prior-art.md). *Library* (1007 references, 11 queries on deltahedra, 4D polytope projections and sections, 600-cell, Platonic face pyramids, space filling): no deltahedron literature in the library. *Web*: Tsuruta et al., ICGG 2014, deltahedral graphs (cgg.cs.tsukuba.ac.jp); Wikipedia, "600-cell", "Deltahedron", "Excavated dodecahedron"; the square bipyramidal honeycomb (via search); Polytope Wiki; MathWorld. Coxeter, *Regular Polytopes* (3rd ed. 1973) is cited for its well-known contents; it was not re-read for this paper.

Data and code. Studies 2026-10-27-4d-shadows (q5.py exact arithmetic, polytopes.py, shadows.py, classify.py, attach.py, shadow17_j11.py, count_exact.py, solve_sage.py; data/shadows.json, all-triangle-matches.json, attachability.json, large-slices-systems.json, form_V48_full_36.json) and 2026-10-27-five-pyramids. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO, Every Mix of the Platonic Solids Fills Space (#47), DICTO DOI 10.5281/zenodo.23551326; DICTO, Aristotle's Pyramid (#44), DICTO DOI 10.5281/zenodo.23529497.

Chapter 5. The White Queen 600-cell (#63, DICTO DOI 10.5281/zenodo.23690856)

Sources checked (27 October 2026; Boole Stott 1900 read 28 October 2026 in the Internet Archive scan of the Verhandelingen, volume 7, and added to the reference library). *History*: J. C. Lagarias and C. Zong, "Mysteries in packing regular tetrahedra", Notices of the AMS 59 (2012), as read for #44 (21 October 2026), for Aristotle, his commentators and Bradwardine; Kepler, *Mysterium Cosmographicum* (1596), and Schläfli's discovery of the 600-cell are cited for their well-known contents, not re-read. *Web*: Wikipedia, "Alicia Boole Stott"; G. W. Hart, "Models of sections of the 600-cell, made by Alicia Boole Stott" (math.stonybrook.edu); search for the 1900 paper (Proofwiki, MacTutor); Wikipedia, "600-cell"; Wolfram Demonstrations,

"Sections of the 16-Cell"; arXiv 1912.06156, "The Geometry of H4 Polytopes" (title only); KNAW digital library (dwc.knaw.nl, title page only). *Library*: the Dodgson texts (Gutenberg) for every quotation, checked word for word. Coxeter, *Regular Polytopes* (3rd ed. 1973), cited for its well-known contents, not re-read.

Version 3 (28 October 2026): following an independent review, §2 gives the four direction vectors and lists the exact checks; §8 reports Alicia Boole Stott's 1900 paper, now read; the Appendix tabulates all 62 slice patterns; the study files are attached to this record; the wording of DICTO's idea is corrected ("proposed").

Version 2 (28 October 2026): §6 adds DICTO's observation that the fifteen Cleo shells read the same backwards, with mirrored angles adding up to 180°.

Data and code. The study files are attached to this record as a zip file (white-queen-600-study): scripts, exact data and the table of the Appendix with edge lengths and dihedral angles (data/atlas600.json). Study 2026-10-27-white-queen-600 (atlas600.py, render_atlas.py; data/atlas600.json, atlas600.pkl), using the exact modules of study 2026-10-27-4d-shadows (q5.py, polytopes.py, shadows.py) and its Slice-48 data (form_V48_full_36.json). References: DICTO, Shadows of the Fourth Dimension (#62), DICTO DOI 10.5281/zenodo.23687223; DICTO, AXE, CLEO and FUJI: the Gap Pieces of the Icosahedron (#19), DICTO DOI 10.5281/zenodo.23351291. Further study files: squares_midcuts.py, front_cells_exact.py, shadow_turnings_exact.py, cleo_shells.py (data/front_cells_exact.json, shadow_turnings_exact.json, cleo_shells.json). Quotations: L. Carroll, *Alice's Adventures in Wonderland* (1865), *Through the Looking-Glass* (1871), Project Gutenberg texts.

Chapter 6. The Red Queen 120-cell (#65, DICTO DOI 10.5281/zenodo.23716140)

Sources checked (28 October 2026; for Version 2, A. Boole Stott 1900, read in the Internet Archive scan of the Verhandelingen, volume 7, and added to the reference library). Reference library search ("120-cell layers round one cell; Boole Stott sections; Schlegel diagram; 600-cell and 120-cell duality"): H. S. M. Coxeter, *Regular Polytopes*, 3rd ed. (1973); M. Koca, N. Koca, M. Al-Ajmi, branching of the W(H4) polytopes (2011); M. Koca, M. Al-Ajmi, N. Koca, grand antiprism by quaternions (2009); Kaida Shi, overall shapes of the 120-cell and 600-cell (2009); A. Garrison and R. Scott, small covers of the dodecahedron and the 120-cell (2001); J. C. Lagarias and C. Zong, "Mysteries in packing regular tetrahedra", Notices of the AMS (2012). Web search "120-cell layers of cells around one dodecahedron 1, 12, 20, 12, 30": Wikipedia "120-cell" (citing J. Stillwell, "The story of the 120-cell", Notices of the AMS, 2001); S. Schleimer and H. Segerman, "Puzzling the 120-cell", arXiv 1310.3549, and their sculpture talks; the Zometool 120-cell model page of the Ruhr-Universität Bochum. Hess (1883) and the hyperbolic {3,5,3} are cited from Coxeter's book as standard; not read for this paper.

Version 2 (28 October 2026): following an independent review of #63, §6 gives the four direction vectors and the exact checks, §7 reports Alicia Boole Stott's 1900 paper, now read, the Appendix tabulates all 124 slice patterns, and the study files are stated to be attached to this record.

Data and code. The study files are attached to this record as a zip file (red-queen-120-study): scripts, exact data and the table of the Appendix with edge lengths and dihedral angles (data/atlas120.json). Study 2026-10-27-white-queen-polar (red_queen.py, five_pyramids.py, atlas120.py, render_red_queen.py, render_five_pyramids.py, render_atlas120.py; data red_queen.json, five_pyramids.json, atlas120.json), built on the exact source model of #63. References: DICTO, The White Queen 600-cell (#63, DOI 10.5281/zenodo.23690856); Aristotle's Pyramid (#44); the Cleo-Icos (#19).

Papers referred to, not in this book

#20	The FUJI Family	10.5281/zenodo.23357896
#42	The DICTO Sunstar	10.5281/zenodo.23497702
#47	Every Mix of the Platonic Solids Fills Space	10.5281/zenodo.23551326

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About the author

DICTO is an artist in Osaka, Japan. His background is architectural sculpture, stonecarving and Buddhist statuary. He trained as an architectural stonecarver at the City and Guilds of London Art School (1998–2001), worked at St James's Palace, and recarved the Roman lettering of the Civil Service Rifles War Memorial at Somerset House, London, for its rededication in 2002. Since 2009 he has worked in Japan in the sculpture and restoration of Buddhist statues, among them a statue at Takaosan Jingo-ji, Kyoto, traditionally attributed to Kōbō Daishi Kūkai.

He builds geometric sculpture from recycled PET bottles with two modular systems of his own, B1NR and DICTOspheres, joined without glue. He holds six Japanese design registrations; five rest on spherical projections of the Platonic solids, the five forms behind his name: Dodecahedron, Icosahedron, Cube, Tetrahedron, Octahedron. His sculptures have been shown in Kyoto at the Kyoto Municipal Museum of Art (Interdependent Origination III, 2025), the Insho Domoto Museum of Fine Arts, the Kyoto Botanical Garden, the Kansho Exhibition and the Kyoto Takasegawa Art Festival.

The geometry behind the sculptures led to seventy-five papers, gathered in *DICTO: Collected Geometric Works*, and to three books: *HexOct: The Laws of Stacking*, *The Golden Cycle*, and this one, *Alicia in Wonderland: The Red & White Queens* (all 2026). Every paper and book is published openly with its own DICTO DOI at Zenodo. More at https://dictospheres.com.